

SOFTWARE REQUIREMENTS SPECIFICATION

Manufacturing ERP System

A single, self-hosted ERP platform for eleven manufacturing industries

Plastic Products

PVC Pipes

Packaging Materials

Rubber Products

Furniture

Electrical Components

Auto Parts

Steel Products

Aluminum Products

Textile

Chemical

Document

Software Requirements Specification (SRS)

Version

1.0 — Baseline

Platform

Laravel 13 · PHP 8.3+ · MySQL 8 · On-Premise / cPanel

Licensing model

Non-SaaS — one installation per customer

Integrations

WhatsApp Cloud API · Google Firebase (FCM)

Prepared by

Keplersoft

Document Control

How to read this document

This document is written in plain English on purpose. It is meant to be read by four different kinds of people, and each of them needs something different from it:

Reader	What they should read	Why
Business owner / factory owner	Sections 1, 2, 4, 12 and the "How this module works" part of each module	To confirm the software matches how the factory actually runs
Project manager	Sections 2, 10, 16, 17	To plan phases, effort and acceptance
Developer	Everything — especially field tables, validations, Sections 3, 5–9, 11, 13, 15	To build exactly what is specified, with no guessing
Tester / QA	User stories, field tables, validation rules, status flows	Every user story and every validation rule is a test case

Terms used in this document

Term	Meaning in simple words
ERP	Enterprise Resource Planning — one software that runs sales, purchase, stock, production, quality, accounts and people together.
Non-SaaS	Each customer gets their own separate copy of the software and their own database. Nothing is shared between customers.
Module	One functional area of the ERP, for example "Purchase" or "Quality Control".
Master data	Information that is entered once and used many times — customers, suppliers, items, machines.
Transaction	A business event that happens on a date — a sales order, a stock issue, a payment.
Voucher / Document	Any numbered record the user creates, such as PO-2026-00042.
BOM	Bill of Materials — the recipe of a finished product: which raw materials, how much of each.
Work Order	An instruction to the factory to produce a specific quantity of a specific item.
GRN	Goods Receipt Note — the record created when material physically arrives from a supplier.
UOM	Unit of Measure — Kg, Metre, Piece, Litre, Roll, Bundle.
RBAC	Role Based Access Control — users get roles, roles get permissions.
FCM	Firestore Cloud Messaging — Google's service for sending push notifications to phones and browsers.
Audit trail	A permanent record of who changed what, when, and what the old and new values were.

Revision history

Version	Date	Author	Change summary
0.1	Draft	Keplersoft	Initial module list and scope
1.0	23 July 2026	Keplersoft	Baseline release — full functional and non-functional specification

Approval

Role	Name	Signature	Date
Business Owner			
Project Manager			
Technical Lead			
QA Lead			

Table of Contents

1. Introduction	6
2. Product Overview and Scope	8
2.3 User Classes	8
3. System Architecture	10
4. Industry Configuration Layer	12
5. Access Control — RBAC and Authentication	16
6. Activity Logging and Audit Trail	19
7. Notification Engine — WhatsApp and Firebase	21
8. Document and PDF Generation Engine	27
9. Customization Framework	29
10. Functional Modules	30
M01 — Company Setup and Master Data	31
M02 — User, Role and Permission Management	33
M03 — Item Master, UOM, Batch and Serial	35
M04 — Bill of Materials and Routing	37
M05 — CRM and Lead Management	39
M06 — Sales (Quotation to Invoice)	41
M07 — Purchase and Vendor Management	44
M08 — Inventory and Warehouse	46
M09 — Production Planning and Execution	48
M10 — Quality Control	51
M11 — Machine, Mould and Die Maintenance	53
M12 — Dispatch, Packing and Logistics	54
M13 — Accounts, Finance and GST	55
M14 — HR, Attendance and Payroll	58
M15 — Reports, Dashboards and Analytics	59
M16 — Settings, Customization and Utilities	60
M17 — Packing Units, QR Codes and Scan Operations	61
11. Data Model	71
12. End-to-End Business Flows	73
13. Global Validation Standards	74
14. Non-Functional Requirements	76
15. cPanel Deployment Specification	78
16. Implementation Roadmap	82

17. Appendices

1. Introduction

1.1 Purpose of this document

This document describes, in complete detail, what the Manufacturing ERP System must do and how it must behave. It is the single agreed reference for everyone building, testing, buying or approving the system. If something is not written here, it is not part of version 1.0 and must go through a change request.

The document answers four questions for every part of the system: **what does the user want to do** (user stories), **what data does the system need** (field tables), **what is not allowed** (validation rules), and **what happens step by step** (process flows and diagrams).

1.2 The business problem

Small and medium manufacturers in India typically run their factory on a mixture of paper registers, Excel sheets, WhatsApp messages and a basic accounting package. This causes five recurring problems:

1. **Nobody knows the true stock.** Physical stock and book stock differ because issues to production are not recorded on time.
2. **Orders slip.** A sales order is accepted without checking whether raw material or machine capacity is available.
3. **Costing is guesswork.** The real cost of a produced item — material plus scrap plus labour plus machine hours — is never calculated.
4. **No traceability.** When a customer complains about a batch, nobody can trace which raw material lot, which machine and which operator produced it.
5. **Information does not reach people.** The store keeper does not know a dispatch is due today; the owner does not know a payment is overdue.

This ERP solves all five, and pushes the fifth one out to people's phones through WhatsApp and push notifications instead of waiting for them to log in.

1.3 Why one product for eleven industries

Plastic moulding, PVC pipe extrusion, corrugated box making, rubber compounding, furniture assembly, electrical component assembly, auto part machining, steel fabrication, aluminium extrusion, textile processing and chemical batch production look very different on the shop floor. But underneath, all eleven follow the same shape:

Buy raw material → Store it → Convert it using a recipe and a machine → Check quality → Store the finished goods → Dispatch and invoice

The differences are in the *details*: a plastic moulder cares about mould cavity count and cycle time; a pipe extruder cares about running metres and coil length; a chemical plant cares about batch potency and expiry; a textile mill cares about shade lots and shrinkage. These are **configuration differences, not code differences**. Section 4 describes the Industry Configuration Layer that turns one codebase into eleven industry-fit products.

1.4 Scope of version 1.0

In scope	Out of scope (future versions)
All 16 functional modules listed in Section 10	Multi-company consolidation across separate legal entities
Single company, unlimited branches, warehouses and users	Native iOS / Android app (a responsive web app is provided; a Flutter client is a Phase 4 item)

In scope	Out of scope (future versions)
Role based access control with field-level and record-level rules	Machine-level IoT data capture (PLC/SCADA integration)
Full activity log and audit trail on every table	Advanced finite capacity scheduling with optimisation algorithms
WhatsApp Cloud API and Firebase push notification alerts	Automatic e-invoice IRN generation through GSP (hooks are provided, the GSP contract is separate)
Customisable PDF documents and print templates	Barcode hardware supply — the system prints and reads barcodes but hardware is bought by the customer
GST-compliant sales, purchase and accounting	Payroll statutory filing (returns are exported, not e-filed)
Deployment on shared cPanel hosting or a private VPS	Kubernetes / container orchestration

1.5 Success criteria

Goal	How we will know it is achieved
Stock accuracy	Physical stock count matches system stock within 2% for 95% of items after three months of use
Traceability	Any finished goods batch can be traced back to its raw material lots, machine, operator and shift within 30 seconds using one screen
Alert reach	90% of critical alerts are delivered to WhatsApp or push within 60 seconds of the triggering event
Accountability	100% of create, update, delete and approve actions are recorded in the activity log with old and new values
Deployability	A fresh installation on standard cPanel shared hosting completes in under 60 minutes using the supplied installer

2. Product Overview and Scope

2.1 Product perspective — what "non-SaaS" means here

This is a **single-tenant, self-hosted product**. Every customer receives their own copy of the application files and their own MySQL database, installed either on their own server or on their own cPanel hosting account. There is no shared database, no shared storage bucket and no tenant identifier column anywhere in the schema.

This choice has direct consequences on the design, and they must be respected throughout development:

Consequence	What it means for the build
No <code>tenant_id</code> anywhere	Data isolation is achieved by separate databases. Never add a tenant column "just in case" — it adds complexity with zero benefit.
Licence enforcement is offline-friendly	A signed licence key file is placed in <code>storage/app/licence.key</code> . It records customer name, allowed modules, allowed user count and expiry. The system verifies the signature locally; an internet connection is optional.
Upgrades are per-installation	A built-in updater downloads a versioned package, backs up the database, runs migrations and clears caches. Every customer may sit on a different version, so migrations must always be backward safe.
The customer owns the data	A one-click full backup (database dump plus the uploads folder) must be available from inside the application, and via a scheduled cron job.
Performance target is modest hardware	The system must work acceptably on shared hosting with 1 GB PHP memory limit, no Redis and no root access. See Section 15.

2.2 User classes

The following user classes exist by default. Each is a Role in the RBAC system, and each can be renamed, copied or restricted further by the customer.

Role	Who they are	What they mainly do in the system
Super Admin	The implementer or the owner's IT person	Full access, including settings, user creation, licence, backups, integration keys. Cannot be deleted; there must always be at least one.
Management	Owner, Director	Read-only across everything plus dashboards, approvals above value limits, and profitability reports.
Sales Executive	Field or inside sales	Leads, quotations, sales orders, customer follow-ups, order status tracking. Sees only their own customers unless given wider scope.
Sales Manager	Head of sales	Everything a sales executive does, plus discount approval, price list changes and team-wide visibility.
Purchase Officer	Buyer	Purchase indents, RFQs, comparative statements, purchase orders, vendor rating.
Store Keeper	Stores / warehouse in-charge	Goods receipt, material issue to production, stock transfers, physical stock count, bin management.
Production Planner	PPC	Production plans, work orders, machine loading, material requirement planning.
Shop Floor Operator	Machine operator / supervisor	Records production output, rejection, downtime and shift entries against a work order. Very restricted screen set.
QC Inspector	Quality department	Incoming, in-process and final inspections; accept, reject, rework or accept-with-deviation decisions.

Role	Who they are	What they mainly do in the system
Dispatch Officer	Dispatch / logistics	Packing lists, delivery challans, e-way bills, vehicle and transporter details.
Accountant	Accounts department	Invoices, payments, receipts, journal entries, GST returns, ledgers, outstanding reports.
HR Officer	HR / admin	Employees, attendance, leave, shifts, salary processing.
Maintenance Engineer	Maintenance	Machine, mould and die registers, preventive maintenance schedules, breakdown records.
Customer (Portal)	External — optional	Views only their own orders, dispatch status, invoices and outstanding balance.

2.3 Operating environment

Server	Linux (CentOS / AlmaLinux / Ubuntu), Apache or LiteSpeed, cPanel or WHM optional
PHP	8.3 or 8.4, with extensions: bcmath, ctype, curl, dom, fileinfo, gd, intl, mbstring, openssl, pdo_mysql, tokenizer, xml, zip, exif
Framework	Laravel 13 (LTS-track), Blade + Livewire/Alpine for interactive screens
Database	MySQL 8.0+ or MariaDB 10.6+, InnoDB, utf8mb4_unicode_ci
Queue	Database driver by default (works on shared hosting); Redis driver when available
Cache / Session	File or database by default; Redis when available
Client browsers	Chrome, Edge, Firefox, Safari — last two major versions. Minimum screen 1280×720 for desk roles, fully responsive for shop-floor and mobile roles.
External services	WhatsApp Cloud API (Meta Graph API v20+), Google Firebase Cloud Messaging (HTTP v1)

3. System Architecture

3.1 Layered architecture

The application is built in clean layers. Each layer only talks to the layer directly below it. This keeps the code testable and makes it possible to add a mobile API later without rewriting business logic.

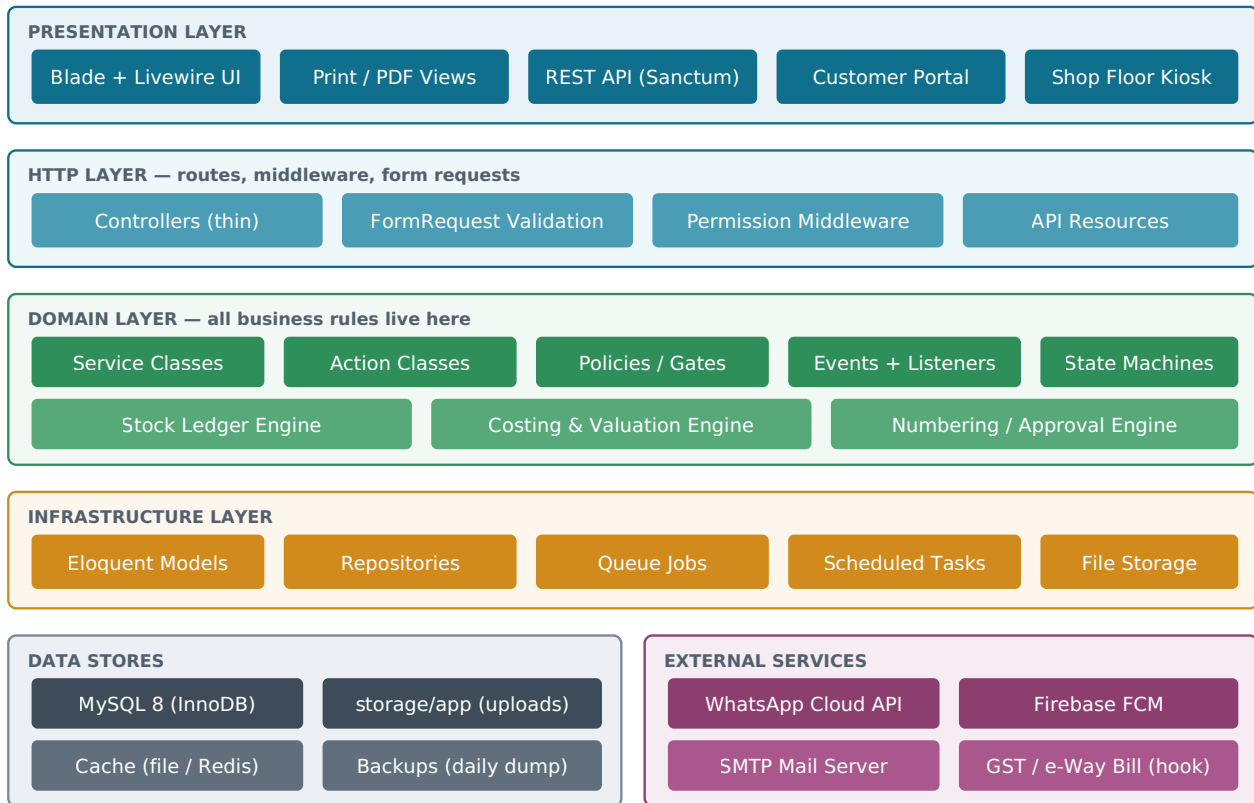


Figure 3.1 — Layered application architecture. Business rules never live in controllers or Blade views.

3.2 Architectural rules the team must follow

#	Rule	Reason
A1	Controllers must not exceed roughly 30 lines per method. They validate, call a service, and return a response.	Keeps logic testable and reusable by both the web UI and the API.
A2	Every write that touches more than one table runs inside a database transaction.	A half-posted GRN that updates stock but not the ledger is the worst kind of bug.
A3	Stock quantity is never stored as a single editable number. It is always derived from the stock ledger, with an optional cached balance table refreshed inside the same transaction.	Makes stock provably correct and auditable.
A4	All money and quantity columns use <code>DECIMAL</code> , never <code>FLOAT</code> . Money is <code>DECIMAL(18,2)</code> , quantity <code>DECIMAL(18,4)</code> , rate <code>DECIMAL(18,4)</code> , percentage <code>DECIMAL(9,4)</code> .	Floating point rounding errors in accounting are unacceptable.

#	Rule	Reason
A5	Nothing is hard deleted. Every table uses soft deletes plus a cancellation reason where relevant.	Audit and statutory record keeping.
A6	Anything that talks to the outside world (WhatsApp, FCM, email) is dispatched to a queue, never called inline in a request.	A slow Meta API must never make the user's Save button hang.
A7	Every document number comes from the central Numbering Engine, using a database lock. Numbers are never generated with <code>max(id)+1</code> .	Prevents duplicate numbers when two users save at the same second.
A8	No raw SQL string concatenation with user input. Query builder or bound parameters only.	SQL injection prevention.
A9	All list screens must be server-side paginated and indexed. No screen may load more than 100 rows at once by default.	Shared hosting has limited memory.
A10	Industry-specific behaviour is driven by configuration, never by <code>if (\$industry == 'plastic')</code> inside a service.	Otherwise the codebase becomes eleven products glued together.

3.3 Module map

The diagram below shows how modules depend on each other. An arrow means "needs data from". Modules with no incoming arrows are foundations and must be built first.

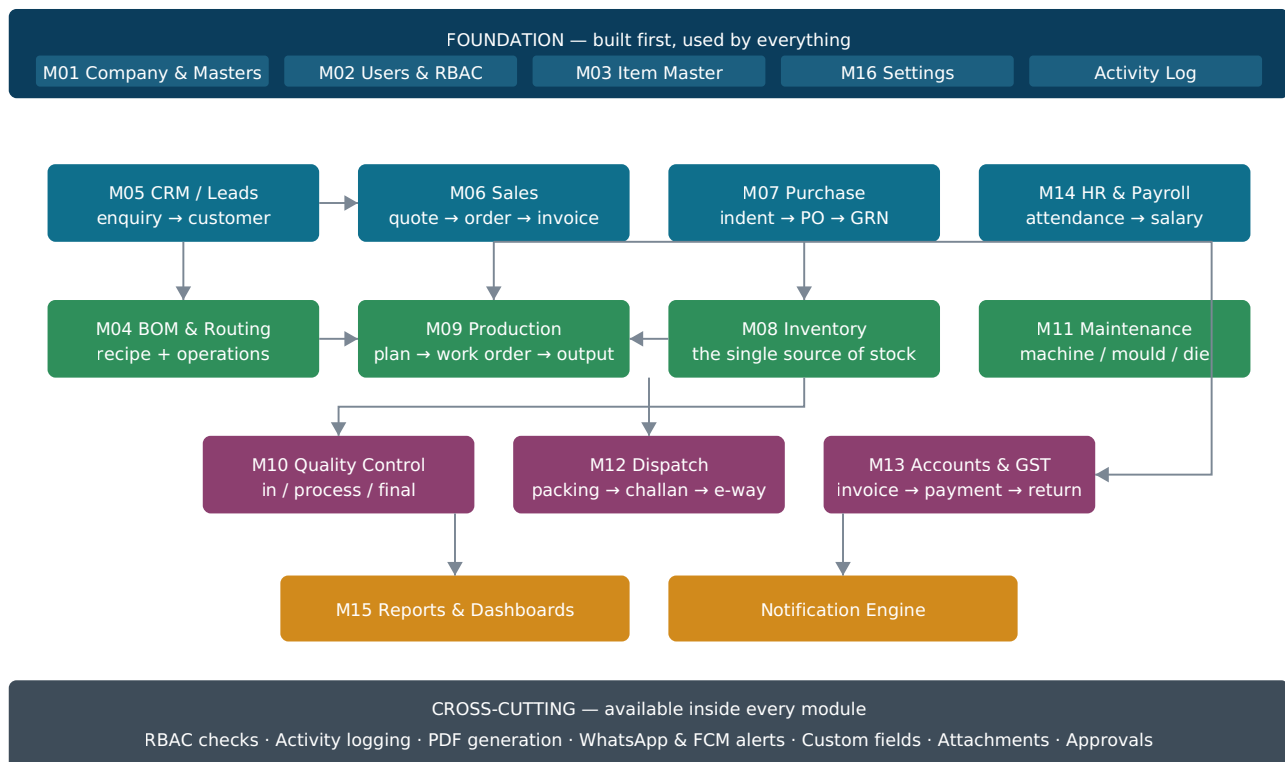


Figure 3.2 — Module dependency map. Build order runs from top to bottom.

4. Industry Configuration Layer

4.1 The idea in one paragraph

Instead of building eleven products, we build one product with an **Industry Profile**. During installation the implementer picks the industry. The profile then switches on the right features, loads the right default masters, applies the right costing method, adds the right extra fields to the item master, and installs the right quality parameters and report set. Everything else in the code stays identical.

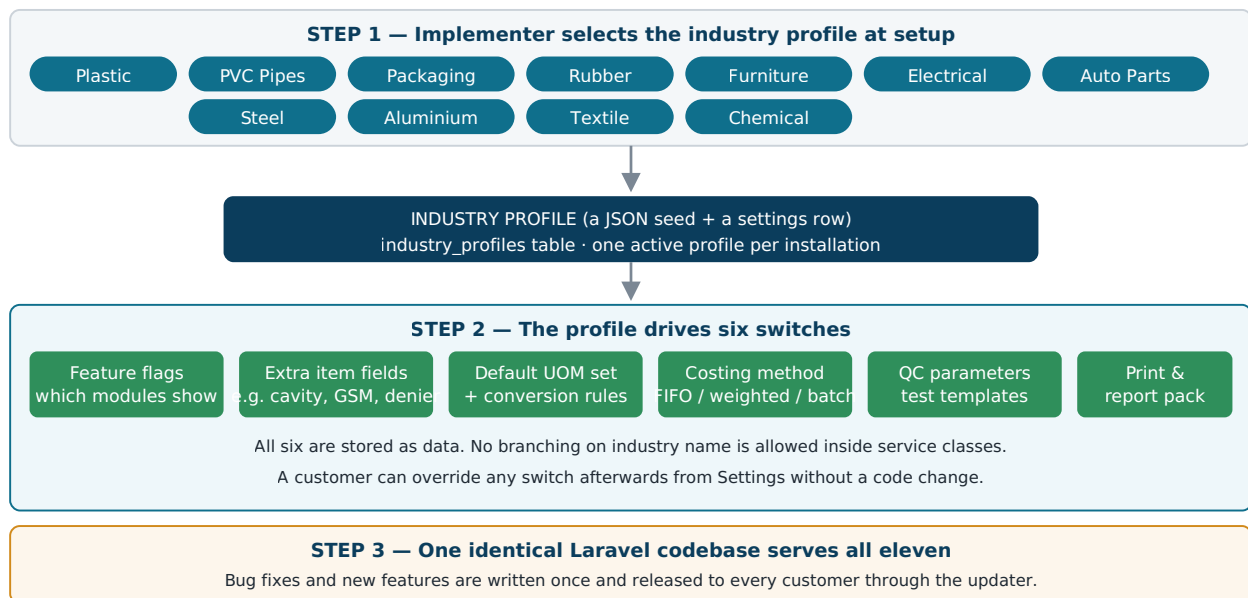


Figure 4.1 — How one codebase becomes eleven industry-fit products.

4.2 What actually differs between the eleven industries

This table is the reference the developer uses when writing the profile seeders. It is deliberately specific, because vague requirements here cause the biggest rework later.

Industry	Primary UOM & second UOM	Item attributes needed	Traceability level	Special production behaviour
Plastic Products	Piece / Kg	Mould code, cavity count, cycle time (sec), shot weight (g), material grade, colour masterbatch %	Batch + machine + mould	Output per shift is calculated from cycle time × cavities. Re grind material re-entry into stock must be recorded as a separate item.
PVC Pipes	Metre / Kg	Outer diameter, wall thickness, pressure class (PN), length per piece, IS standard, colour	Batch + extruder line + shift	Continuous extrusion. Production is entered in running metres and converted to Kg using metre-per-Kg factor. Off-cut scrap is common and must be tracked.
Packaging Materials	Piece / Sq.m / Kg	Ply count, GSM, flute type, board size L×W×H, printing colours, bursting strength	Batch + reel lot	Reel-to-sheet conversion with a wastage percentage. Job-work (printing at an outside vendor) is frequent.

Industry	Primary UOM & second UOM	Item attributes needed	Traceability level	Special production behaviour
Rubber Products	Piece / Kg	Compound code, hardness (Shore A), curing time and temperature, mould code, ageing/expiry of compound	Batch (compound lot) mandatory	Two-stage: compound mixing produces an intermediate item with a shelf life, then moulding consumes it. Expiry-based FEFO issue is mandatory.
Furniture	Piece / Set	Model, dimensions, material (ply/MDF/steel), finish, colour, upholstery code, knock-down or assembled	Serial number for high value items	Multi-level BOM with sub-assemblies. Made-to-order variants are common, so BOM must support per-order customisation.
Electrical Components	Piece / Box	Voltage rating, current rating, IP rating, certification (ISI/CE), pin type, wire gauge	Serial number + batch	Assembly BOM with many small purchased parts. Serial numbers are needed for warranty claims.
Auto Parts	Piece / Set	Part number, OEM part number, vehicle model applicability, drawing revision, material grade, tolerance class	Serial or lot, plus drawing revision — mandatory	Customer schedules (weekly/monthly delivery schedules) instead of one-off orders. PPAP / control-plan documents must attach to the item.
Steel Products	Kg / Piece / Metre	Grade, thickness, width, length, heat number, coating, IS/ASTM standard	Heat number is mandatory and must flow from raw material to finished goods	Weight is calculated from dimensions using a density formula. Cutting plans generate off-cuts that must return to stock as usable remnants.
Aluminium Products	Kg / Metre	Alloy and temper (e.g. 6063-T6), section code, die code, weight per metre, anodising/powder-coat spec	Batch + die + billet lot	Extrusion output is measured in metres and billed in Kg. Die life tracking (number of extrusions per die) is required.
Textile	Metre / Kg / Roll	Count, GSM, width, weave, shade/lot number, shrinkage %, fibre composition	Shade lot is mandatory — mixing lots causes customer rejection	Heavy job-work (dyeing, printing at outside processors). Grey-to-finished conversion loses length and gains/loses weight; both must be recorded.
Chemical	Kg / Litre	Concentration/assay %, density, hazard class, MSDS reference, shelf life, storage temperature	Batch + expiry, mandatory, with full genealogy	Batch process manufacturing with variable yield. Potency-based issue (issue more of a weaker lot). Expiry-driven FEFO. Retention samples must be logged.

4.3 The industry profile record

One row in `industry_profiles`, plus a JSON configuration column. Below is the shape of the configuration.

```
{
  "code": "pvc_pipes",
  "name": "PVC Pipes & Fittings",
  "modules": {
    "production": true, "quality": true, "maintenance": true,
    "jobwork": false, "batch_tracking": true, "serial_tracking": false,
    "mould_die_register": true, "shade_lot": false, "heat_number": false
  },
  "uom": {
    "default_stock_uom": "MTR",
    "allow_dual_uom": true,
    "secondary_uom": "KG",
    "conversion_mode": "item_specific_factor"
  },
  "costing": { "method": "weighted_average", "include_scrap_credit": true },
  "item_attributes": [
    { "key": "outer_diameter_mm", "label": "Outer Diameter (mm)", "type": "decimal", "required": true,
      "min": 1, "max": 2000 },
    { "key": "wall_thickness_mm", "label": "Wall Thickness (mm)", "type": "decimal", "required": true,
      "min": 0.1, "max": 200 },
    { "key": "pressure_class", "label": "Pressure Class", "type": "select", "options":
      ["PN4", "PN6", "PN10", "PN16"], "required": true },
    { "key": "length_per_piece_m", "label": "Length per Piece (m)", "type": "decimal", "required": true,
      "min": 0.1 },
    { "key": "is_standard", "label": "IS Standard", "type": "text", "required": false,
      "max_length": 40 }
  ],
  "qc_templates": ["incoming_resin", "inprocess_extrusion", "final_pipe_dimension",
    "hydraulic_pressure_test"],
  "reports": ["metre_wise_production", "line_efficiency", "scrap_percentage", "extruder_downtime"],
  "print_templates": ["pipe_bundle_label", "tax_invoice_with_metres"]
}
```

Design rule. The profile is loaded once per request and cached. Services ask questions like `Industry::feature('batch_tracking')` or `Industry::costingMethod()`. They never ask `Industry::code() === 'pvc_pipes'`. If a new industry needs behaviour that no existing flag expresses, add a new flag — do not add an `if` on the industry name.

4.4 Dual unit of measure — the most common source of confusion

Several of these industries buy in one unit and sell in another. A pipe factory buys resin in Kg, produces pipes measured in metres, and invoices the customer in metres but computes freight on Kg. The system handles this with three ideas:

Concept	Meaning and rule
Stock UOM	The one unit in which the stock ledger stores quantity for that item. It can never be changed once a transaction exists for the item. Validation must block the change.
Transaction UOM	The unit the user types on a document. Any UOM linked to the item is allowed. The system converts to stock UOM before writing the ledger.
Conversion factor	Stored per item, not globally, because 1 metre of 110 mm pipe is not the same weight as 1 metre of 25 mm pipe. Factor is <code>DECIMAL(18,6)</code> and must be greater than zero.

Both quantities are stored on every transaction line: `qty` in the transaction UOM and `base_qty` in the stock UOM. Reports can then show either. Rounding is applied only at display time, never before the ledger is written.

4.5 Costing methods by industry

Method	Suits	How it works
Weighted Average	PVC pipes, plastic, aluminium, steel, packaging	Every receipt recalculates the average rate: $(\text{old value} + \text{new value}) \div (\text{old qty} + \text{new qty})$. Simple and stable when raw material prices move gradually.
FIFO	Furniture, electrical, auto parts	The oldest received cost layer is consumed first. Gives an accurate cost when prices move sharply, and needs a cost-layer table.
Batch / Specific Cost	Chemical, rubber, textile	Each batch carries its own cost. Issue picks a specific batch, so the exact cost of that batch is consumed. Mandatory when batches differ in potency or shade.
Standard Cost	Optional add-on for auto parts	A frozen planned cost is used for valuation and the difference from actual is posted as a variance. Used when the customer runs formal cost accounting.

Important. The costing method is chosen at installation and locked once the first financial year closes. Changing it later requires a controlled re-valuation utility with an audit entry, available only to Super Admin.

5. Access Control — RBAC and Authentication

5.1 The model in plain words

A **user** is a person who logs in. A **role** is a job title such as Store Keeper. A **permission** is one specific thing that can be done, such as "approve a purchase order". Permissions are given to roles, roles are given to users. A user may hold more than one role, and may also be given or denied a single permission directly, which overrides the role.

On top of this the system adds two more layers that most small ERPs miss:

- **Record scope** — *which* records the user may see. A sales executive with the "view sales order" permission still only sees orders of customers assigned to them.
- **Field rules** — *which columns* the user may see or edit. A production planner may open a sales order but must not see the selling price.

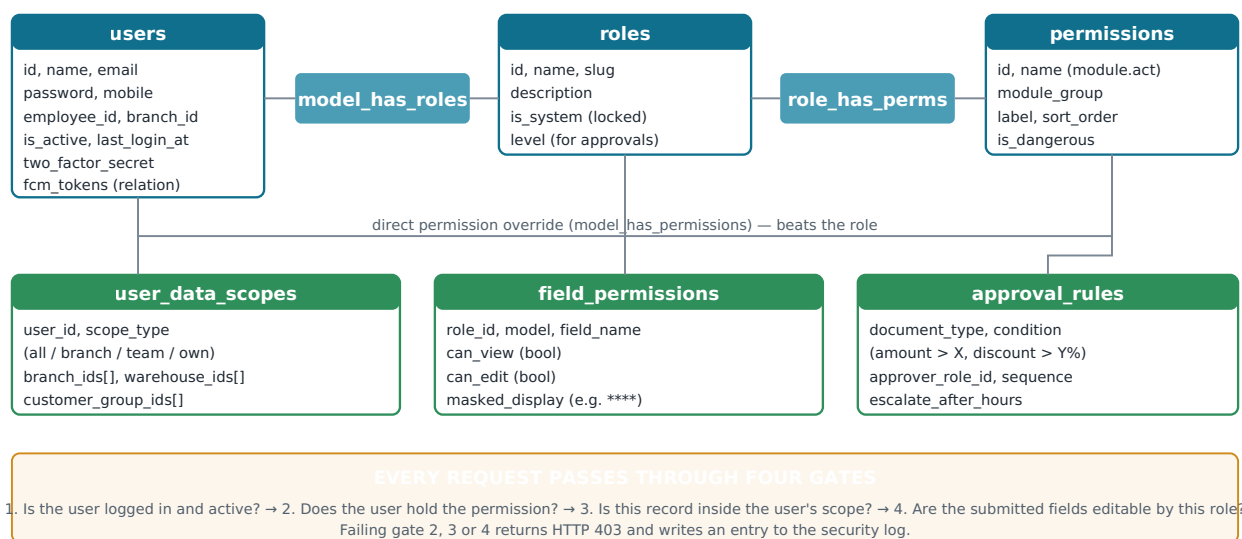


Figure 5.1 — Access control data model and the four gates every request passes.

5.2 Permission naming convention

Every permission is named `module.action` in lower snake case. This is not cosmetic — the settings screen groups permissions automatically by the part before the dot, and the seeder generates them from a single array.

Action suffix	Example	Meaning
<code>.view</code>	<code>sales_order.view</code>	Open the list and the detail screen (read only)
<code>.create</code>	<code>sales_order.create</code>	Create a new draft record
<code>.update</code>	<code>sales_order.update</code>	Edit a record that is still editable in its current status
<code>.delete</code>	<code>sales_order.delete</code>	Soft delete a draft record
<code>.approve</code>	<code>sales_order.approve</code>	Move a record from Pending Approval to Approved
<code>.cancel</code>	<code>sales_order.cancel</code>	Cancel a confirmed record, with a mandatory reason
<code>.print</code>	<code>sales_order.print</code>	Generate the PDF
<code>.export</code>	<code>sales_order.export</code>	Download Excel or CSV — treated as a sensitive permission and always logged

Action suffix	Example	Meaning
<code>.view_rate</code>	<code>sales_order.view_rate</code>	See price and value columns. Withholding it hides money from shop-floor roles.
<code>.reopen</code>	<code>work_order.reopen</code>	Return a closed document to an open status. Marked as a dangerous permission.

5.3 User stories — access control

US-RBAC-01 — As a **Super Admin**, I want to create a role and tick permissions on a grouped checklist, so that I can set up a new job function in a few minutes without a developer.

Acceptance: permissions appear grouped by module with a "select all in module" checkbox; saving writes an activity log entry listing added and removed permissions.

US-RBAC-02 — As a **Super Admin**, I want to copy an existing role, so that I can make a slight variation without ticking eighty boxes again.

Acceptance: the copy is created with " (Copy)" appended to the name and identical permissions; the name must then be edited to something unique.

US-RBAC-03 — As a **Sales Manager**, I want my executives to see only their own customers' orders, so that team members do not poach or leak each other's accounts.

Acceptance: with scope = "own", every list query is automatically filtered by `assigned_user_id`; direct URL access to another user's record returns 403 and is logged.

US-RBAC-04 — As an **Owner**, I want purchase orders above ₹2,00,000 to need my approval, so that large spending is controlled.

Acceptance: an approval rule with condition `grand_total > 200000` routes the PO to the Management role; the PO stays in "Pending Approval" and cannot generate a GRN until approved.

US-RBAC-05 — As a **Production Planner**, I want to open a sales order to plan production, but I should not see the selling price.

Acceptance: a field permission on `rate`, `amount` and `grand_total` for the Planner role sets `can_view = false`; those columns are removed from the screen, from the exported file and from the API response — not merely hidden by CSS.

US-RBAC-06 — As a **Super Admin**, I want to deactivate a user who has left the company without deleting them, so that their old documents keep showing their name.

Acceptance: `is_active = false` blocks login immediately and revokes all API tokens and FCM devices, while historical records remain intact.

US-RBAC-07 — As a **Super Admin**, I want to force two-factor authentication for finance roles, so that accounting data is protected.

Acceptance: a per-role flag `require_2fa`; on next login the user must set up a TOTP app before reaching any other screen.

5.4 Authentication requirements

Requirement	Rule
Login identifier	Email or mobile number or username, configurable in Settings. Case-insensitive, trimmed.
Password policy	Minimum 8 characters, at least one letter and one digit. Configurable to require symbols and 12 characters. Checked against a list of the 1000 most common passwords.

Requirement	Rule
Hashing	Bcrypt with cost 12, or Argon2id where the PHP build supports it.
Brute force protection	Maximum 5 failed attempts per email plus IP within 10 minutes, then a 15-minute lockout. Every lockout raises a security alert to Super Admin.
Session	Idle timeout configurable, default 120 minutes. Absolute timeout 12 hours. Session fixation prevented by regenerating the session ID on login.
Concurrent sessions	Optional setting to allow only one active session per user; logging in elsewhere ends the previous session.
Password reset	Signed token valid for 60 minutes, single use. Delivered by email and optionally by WhatsApp. The reset link never contains the user's password or email in plain text in a query string that is logged.
First login	Users created by an admin receive a temporary password and must change it before reaching any screen.
API access	Laravel Sanctum personal access tokens, scoped to permissions, with a visible token list and one-click revoke per token.
Impersonation	Super Admin may "log in as" another user for support. The banner is always visible, the session is time limited to 30 minutes, and every action during impersonation is logged with both user IDs.

6. Activity Logging and Audit Trail

6.1 What must be logged

The requirement is simple to state and demanding to implement: **every action that changes data, and every action that reads sensitive data, must leave a permanent record.** Three separate logs are maintained because they answer different questions and have different retention needs.

Log	Answers the question	Contents	Retention
Activity Log	"Who changed this record and what exactly did they change?"	Model type and ID, event (created/updated/deleted/restored), old values, new values, user, IP, user agent, timestamp, document number for readability	Never purged. Archived to a yearly partition table after 3 years.
Access Log	"Who looked at this, and who tried to look at it?"	Route, method, model viewed, exports and prints, failed permission checks (403), login, logout, failed login, impersonation start/stop	13 months, then aggregated.
System Log	"Why did the integration or job fail?"	Queue job failures, WhatsApp and FCM send results, scheduled task runs, migration runs, backup results, PDF generation errors	90 days.

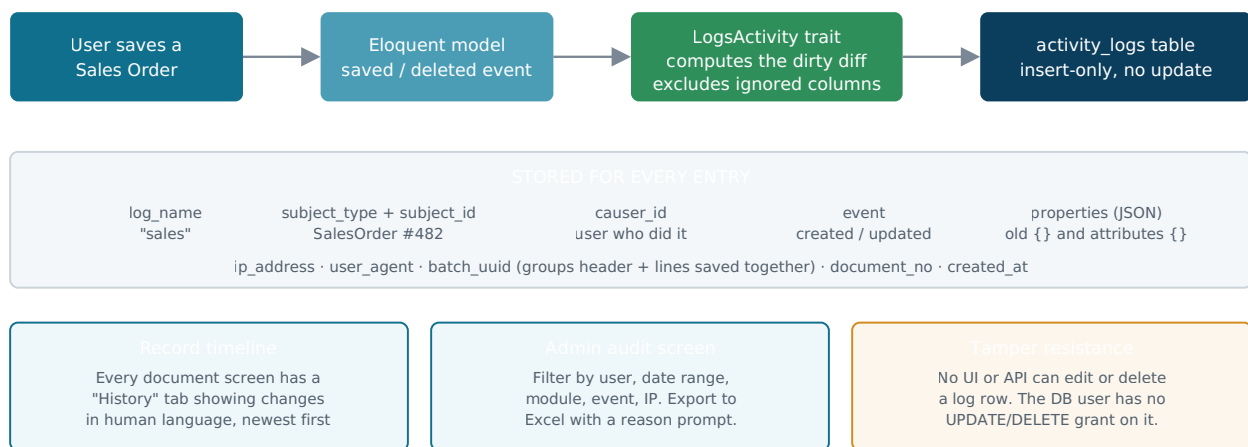


Figure 6.1 — Activity logging pipeline. Logging is automatic through a model trait, not written by hand in each controller.

6.2 Logging rules

#	Rule	Detail
L1	Logging is automatic	A base model trait attaches to model events. A developer adding a new module gets logging for free by extending the base model.
L2	Never log secrets	password, remember_token, two_factor_secret, API keys and access tokens are on a global ignore list. The list is enforced centrally, not per model.
L3	Only the difference is stored	If a user edits a 60-column record and changes one field, the properties column holds one old value and one new value — not sixty.

#	Rule	Detail
L4	Header and lines share a batch	Saving a sales order with 12 lines produces one batch UUID so the timeline shows it as a single business event, expandable into detail.
L5	Business events are logged too	Beyond column changes, named events are recorded: "PO approved", "Invoice emailed to customer", "Stock adjusted after physical count", "WhatsApp alert sent". These carry a readable description.
L6	Exports and prints are logged	Because an exported customer list is the most common data leak, every export records the module, filter used, row count and file name.
L7	Failed permission attempts are logged	Repeated 403s from the same user raise an alert to Super Admin — this usually means either a wrongly configured role or someone probing.
L8	Logs are written in the same transaction	If the business write rolls back, the log rolls back with it. There must never be a log entry for a change that did not happen.
L9	Performance guard	Bulk imports write a single summarised log entry ("Imported 2,340 items from file X") plus per-row logs written by a queued job, so the import itself stays fast.

6.3 User stories — audit

US-LOG-01 — As an **Owner**, I want to see who changed the rate on invoice INV-2026-00187, so that I can find out why the customer was billed differently from the quotation.

Acceptance: the invoice's History tab shows "Rate changed from 112.00 to 108.00 on line 2 (Item: 110mm SWR Pipe) by Ramesh Patel on 14 Jul 2026 at 15:42 from IP 103.x.x.x".

US-LOG-02 — As a **Super Admin**, I want to see everything one user did yesterday, so that I can investigate a complaint.

Acceptance: the audit screen filtered by user and date shows a chronological list across all modules, with links to each affected record.

US-LOG-03 — As an **Auditor**, I want to prove that stock was never edited directly, so that I can rely on the stock ledger.

Acceptance: the system exposes no route that writes a stock balance directly; every balance change traces to a ledger entry, and every ledger entry traces to a source document with an activity log entry.

7. Notification Engine — WhatsApp and Firebase

7.1 Why this matters

In a real factory, people do not sit in front of the ERP. The store keeper is at the gate, the supervisor is on the shop floor, the owner is travelling. An ERP that requires people to log in to discover a problem will not change how the business runs. So the ERP must **push information out** — to WhatsApp for people outside the system (customers, suppliers, transporters) and to push notifications for people inside it (staff with the app or browser open).

7.2 Architecture of the engine

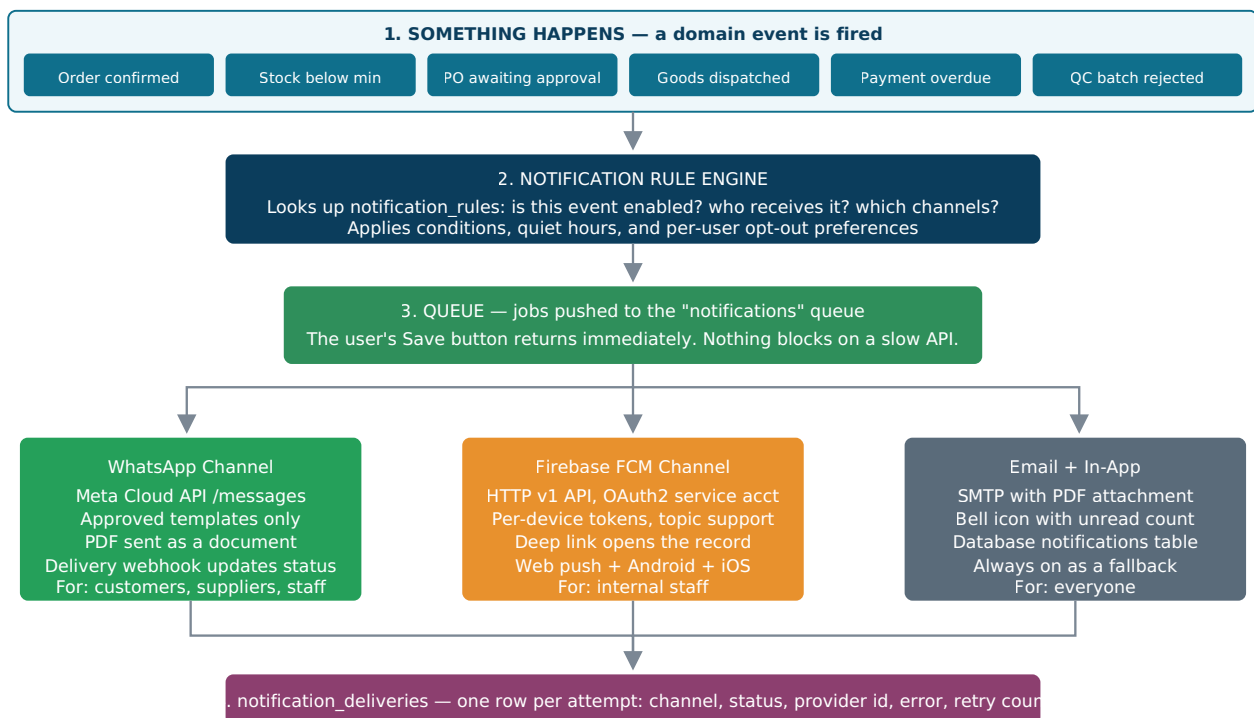


Figure 7.1 — Event to delivery pipeline. Every attempt is recorded so failures are visible, not silent.

7.3 Alert catalogue

These alerts ship with the system. Each one can be switched on or off, re-targeted to different roles, and reworded, from the Settings screen. "W" = WhatsApp, "P" = push, "E" = email, "I" = in-app.

Code	Event	Default recipients	Channels	Message content
N01	Quotation sent	Customer contact	W, E	Quotation number, validity date, total value, PDF attached
N02	Sales order confirmed	Customer, Sales Exec, Production Planner	W, P, I	Order number, items, promised delivery date
N03	Order needs approval (discount or credit limit)	Sales Manager	P, I, W	Order number, customer, discount %, direct approve link
N04	Purchase order approved	Supplier contact, Purchase Officer	W, E	PO number, items, delivery date, PDF attached
N05	Material received (GRN posted)	Purchase Officer, QC Inspector	P, I	GRN number, supplier, items awaiting inspection

Code	Event	Default recipients	Channels	Message content
N06	Stock below reorder level	Purchase Officer, Store Keeper	P, I, W	Item name, current stock, reorder level, suggested order qty
N07	Batch nearing expiry (chemical, rubber)	Store Keeper, QC	P, I	Item, batch, expiry date, remaining quantity and value
N08	Work order released to shop floor	Shop Floor Supervisor	P, I	Work order number, item, quantity, machine, target date
N09	Production target missed for the shift	Production Planner, Management	P, W	Machine, planned vs actual quantity, shift, reason if entered
N10	QC rejection above threshold	QC Head, Production Planner, Management	W, P, I	Item, batch, rejected quantity, rejection percentage, defect reason
N11	Machine breakdown logged	Maintenance Engineer, Production Planner	P, W	Machine code, breakdown time, reported by, affected work orders
N12	Preventive maintenance due	Maintenance Engineer	P, I	Machine or mould code, due date, last service date
N13	Goods dispatched	Customer, Sales Exec	W	Invoice number, vehicle number, transporter, LR number, e-way bill, expected arrival
N14	Invoice generated	Customer, Accountant	W, E	Invoice number, amount, due date, PDF attached
N15	Payment received	Customer, Accountant	W	Receipt number, amount, mode, remaining outstanding balance
N16	Payment overdue reminder	Customer, Sales Exec, Accountant	W, E	Invoice number, days overdue, amount. Scheduled daily at a configurable time.
N17	Credit limit exceeded	Sales Manager, Accountant	P, I	Customer, limit, current outstanding, the order that breached it
N18	Employee absent / late	HR Officer, Department Head	P, I	Employee name, department, punch-in time or absence
N19	Salary slip generated	Employee	W, E	Month, net pay, PDF slip attached
N20	Failed login attempts / new device login	Super Admin, affected user	P, E	User, IP address, time, device
N21	Daily business summary	Management	W, E	Yesterday's sales value, production quantity, dispatches, collections, top pending items. Scheduled at 09:00.
N23	Package scanned out for dispatch	Customer contact	W	Package count, item, batch, vehicle number, expected delivery date
N24	Dispatch scan refused (QC hold, wrong order, already dispatched)	Dispatch Officer, QC Inspector, Sales owner	P, I	Package code, refusal reason, who scanned it, which document it belongs to
N25	Label reprinted more than twice for one package	Super Admin, Store Manager	P, E	Package code, reprint count, users and reasons — a duplicate-label warning
N26	Offline scan queue failed to apply	Store Manager, Super Admin	P, I	Device, operator, number of failed scans, original scan times, first failure reason
N22	Backup completed or failed	Super Admin	E, I	Backup file name, size, duration, or the failure reason

7.4 WhatsApp Cloud API integration

7.4.1 Setup fields (Settings → Integrations → WhatsApp)

Field	Type	Required	Validation and notes
Enable WhatsApp	Toggle	Yes	When off, the channel is skipped silently and only in-app/email are used.
Phone Number ID	Text	Yes	Digits only, 10-20 characters. Taken from the Meta app dashboard.
WhatsApp Business Account ID	Text	Yes	Digits only. Needed to read the template list.
Permanent Access Token	Password	Yes	Stored encrypted with the application key. Never displayed after saving — only "Change" is offered. Never written to any log.
API Version	Select	Yes	Default v20.0. Kept configurable so a version bump does not need a code release.
Webhook Verify Token	Text	Yes	Random string, minimum 20 characters, generated by a button.
Default Country Code	Text	Yes	Default 91. Used to normalise mobile numbers that were saved without a country code.
Sandbox / Test Number	Text	No	When set, all outgoing messages are redirected here instead of the real recipient. Essential during UAT. A red banner shows in the UI while it is active.
Daily send limit	Integer	Yes	Default 1000. Protects against a runaway loop sending thousands of messages.
Quiet hours	Time range	No	Default 21:00-08:00. Non-critical messages are held and released after the window.
Test connection	Button	—	Sends a template message to a number the admin types, and shows the raw API response.

7.4.2 Template management

WhatsApp only allows business-initiated messages using templates approved by Meta. The ERP therefore keeps a local `whatsapp_templates` table that mirrors the approved templates and maps each ERP alert to one of them.

Field	Rule
template_name	Exactly as approved in Meta, lower snake case. Required, unique.
language_code	e.g. <code>en</code> , <code>en_US</code> , <code>gu</code> , <code>hi</code> . Required.
category	UTILITY / MARKETING / AUTHENTICATION. Transactional ERP alerts must be UTILITY.
body_variables	Ordered list mapping <code>{{1}}</code> , <code>{{2}}</code> ... to ERP placeholders such as <code>{customer_name}</code> , <code>{document_no}</code> , <code>{amount}</code> .
has_document_header	Boolean. When true the alert attaches the generated PDF as the header document.
status	APPROVED / PENDING / REJECTED, refreshed by a "Sync templates" button that reads the Meta API.

Common mistake to avoid. Do not build free-text WhatsApp messages for outbound alerts. They will only be delivered inside a 24-hour customer service window and will silently fail otherwise. Always send an approved template, and use free text only when replying inside an open conversation.

7.4.3 Outgoing payload — invoice alert with PDF

```
POST https://graph.facebook.com/v20.0/{PHONE_NUMBER_ID}/messages
Authorization: Bearer {PERMANENT_ACCESS_TOKEN}
Content-Type: application/json

{
  "messaging_product": "whatsapp",
  "recipient_type": "individual",
  "to": "919876543210",
  "type": "template",
  "template": {
    "name": "invoice_generated",
    "language": { "code": "en" },
    "components": [
      { "type": "header",
        "parameters": [
          { "type": "document",
            "document": { "link": "https://erp.customer.com/doc/inv/9f2c8b17-signed",
                          "filename": "INV-2026-00187.pdf" } } ] },
      { "type": "body",
        "parameters": [
          { "type": "text", "text": "Shreeji Traders" },
          { "type": "text", "text": "INV-2026-00187" },
          { "type": "text", "text": "1,24,560.00" },
          { "type": "text", "text": "22-Aug-2026" } ] }
    ]
  }
}
```

The document link must be a **signed, time-limited public URL** generated by the ERP (valid 7 days). It must not require login, must not be guessable, and must be revocable. Meta's servers download the file, so a link behind the ERP login will fail.

7.4.4 Webhook handling

Meta calls back with delivery status. The ERP exposes `GET/POST /api/webhooks/whatsapp`:

- **GET** — verification handshake. Compare `hub.verify_token` with the stored verify token and echo `hub.challenge`.
- **POST** — status updates (`sent`, `delivered`, `read`, `failed`) and inbound replies. The route is exempt from CSRF but must verify the `X-Hub-Signature-256` HMAC using the app secret. Invalid signature returns 403 and is logged.
- Statuses update `notification_deliveries.status` so the ERP can show a tick mark next to "Invoice sent on WhatsApp — delivered 15:42, read 16:03".
- Inbound replies are stored and shown on the customer's timeline, so a "please send updated proforma" reply is not lost.

7.4.5 Failure handling

Situation	Behaviour
Network timeout / HTTP 5xx	Retry with exponential backoff: 1 min, 5 min, 15 min, 1 h, 4 h. Maximum 5 attempts, then mark failed and notify Super Admin in-app.
HTTP 429 rate limited	Respect <code>Retry-After</code> ; requeue the whole notification queue with a delay rather than hammering.
Invalid number (error 131026 or similar)	Do not retry. Mark the contact's WhatsApp number as invalid and flag it on the customer screen so someone corrects it.
Template not approved	Do not retry. Fall back to email if an address exists, and raise an in-app alert to Super Admin naming the template.

Situation	Behaviour
Token expired	Stop the channel, raise a high-priority alert, and show a red banner on the dashboard for Super Admin. Do not keep burning attempts.

7.5 Firebase Cloud Messaging integration

7.5.1 Setup fields (Settings → Integrations → Firebase)

Field	Type	Required	Validation and notes
Enable Push	Toggle	Yes	—
Service Account JSON	File upload	Yes	Must be valid JSON containing <code>project_id</code> , <code>client_email</code> and <code>private_key</code> . Stored outside the public web root, permissions 0600, encrypted at rest.
Project ID	Text (read-only)	Auto	Extracted from the uploaded JSON and shown for confirmation.
Web Push VAPID key	Text	Yes for browser push	From Firebase console → Cloud Messaging → Web configuration.
Web API key, Auth domain, Sender ID, App ID	Text	Yes for browser push	Used by the front-end SDK. Safe to expose in the browser, unlike the service account.
Default notification icon and colour	File / colour	No	Icon 192×192 PNG.
Send test push	Button	—	Sends to the admin's own registered devices and shows the raw response.

7.5.2 Device token lifecycle

Stage	Behaviour
Registration	On login, the browser or app requests permission and posts the FCM token to <code>POST /api/devices</code> with device type, model and app version. Stored in <code>user_devices</code> (<code>user_id</code> , token unique, platform, last_seen_at).
Refresh	FCM rotates tokens. The client posts the new token on every app start; the old row is replaced, matched on the device installation ID.
Cleanup	A response of <code>UNREGISTERED</code> or <code>INVALID_ARGUMENT</code> deletes the token immediately. A weekly job removes tokens unused for 90 days.
Logout / deactivate	Deleting the user or setting <code>is_active = false</code> removes all their device tokens in the same transaction.

7.5.3 Outgoing payload

```
POST https://fcm.googleapis.com/v1/projects/{PROJECT_ID}/messages:send
Authorization: Bearer {OAUTH2_ACCESS_TOKEN – obtained from the service account, cached ~55 min}

{
  "message": {
    "token": "d7Kj...device_token",
    "notification": {
      "title": "Purchase Order needs your approval",
      "body": "PO-2026-00214 · Mahalaxmi Polymers · Rs 3,45,600"
    },
    "data": {
      "type": "purchase_order_approval",
      "module": "purchase",
      "record_id": "214",
      "deep_link": "/purchase-orders/214",
      "priority": "high"
    },
    "android": { "priority": "HIGH",
      "notification": { "channel_id": "erp_approvals", "sound": "default" } },
    "apns": { "payload": { "aps": { "sound": "default", "badge": 1 } } },
    "webpush": { "fcm_options": { "link": "https://erp.customer.com/purchase-orders/214" } }
  }
}
```

Rules: the `data` block always carries `type`, `module`, `record_id` and `deep_link` so tapping the notification opens the exact record. Title must stay under 65 characters and body under 240, because both are truncated on the lock screen. Batch sends use multicast in chunks of 500 tokens.

7.6 Notification preference rules

#	Rule	Detail
NR1	Three levels of control	System default → role default → individual user preference. The most specific one wins.
NR2	Users cannot mute mandatory alerts	Alerts flagged <code>is_mandatory</code> (approvals addressed to them, security alerts) are always delivered in-app even if other channels are muted.
NR3	De-duplication	The same event to the same recipient on the same channel is not sent twice within a configurable window (default 30 minutes), identified by an event fingerprint hash.
NR4	Digest option	High-frequency alerts (low stock, QC rejections) can be set to "digest" so one summary message is sent per hour or per day instead of one per event.
NR5	Customer consent	Each customer and supplier contact has a <code>whatsapp_opt_in</code> flag with the date it was recorded. No WhatsApp message goes to a contact without it.
NR6	Everything is visible	Settings → Notification Log lists every attempt with recipient, channel, status, and the exact rendered message, filterable and exportable.

8. Document and PDF Generation Engine

8.1 Requirements

Every document the factory hands to a customer, supplier or transporter is generated by the ERP. Printed output is often the most visible part of the software to outsiders, so it must be correct, branded and configurable without touching code.

Requirement	Detail
Engine	An HTML-to-PDF renderer driven by Blade templates. Dompdf is the default because it is pure PHP and needs no binary — essential on shared cPanel hosting. Where the server allows a binary, a headless Chromium renderer may be enabled in Settings for better CSS support. Templates must render acceptably on both.
Templates are data	Each document type has a record in <code>print_templates</code> holding the Blade/HTML body, paper size, orientation, margins and a flag for which template is the default. Customers can have several templates for one document type and pick one at print time.
Safe editing	The template editor accepts a restricted syntax — placeholders and simple loops only. Raw PHP is never evaluated. Placeholders are resolved from a whitelisted data array, so a template can never read the database directly.
Live preview	The editor shows a preview rendered with sample data before saving.
Letterhead handling	Three modes: full design printed by the ERP; pre-printed stationery (the ERP leaves the top 45 mm blank, configurable); or a background image on every page.
Multi-copy	Tax invoices print "Original for Recipient", "Duplicate for Transporter", "Triplicate for Supplier" as separate pages or a single page with the label configurable.
Amount in words	Indian numbering (lakh, crore) with configurable currency name. Must handle paise correctly.
Barcode and QR	Code128 and QR generation for labels, e-invoice QR, and item barcodes. Rendered as inline SVG or base64 PNG, never as an external URL.
Page numbering	"Page X of Y" on multi-page documents, with the item table header repeating on each page and a "carried forward" subtotal.
Digital signature image	Optional authorised signatory image and "This is a computer generated document" footer text.
Storage	Generated PDFs for outgoing documents are stored under <code>storage/app/documents/{year}/{type}/</code> with a UUID filename, so the same file can be re-sent without regeneration and the customer sees exactly what was sent.
Performance	A 100-line invoice must render in under 3 seconds. Bulk printing of more than 20 documents is queued and delivered as a ZIP through the notification bell.

8.2 Document types shipped as standard

Sales	Purchase	Stores & Production	Other
Quotation Proforma Invoice Sales Order Confirmation Delivery Challan Tax Invoice Credit / Debit Note Customer Ledger & Outstanding	Purchase Enquiry / RFQ Comparative Statement Purchase Order Goods Receipt Note Purchase Return / Debit Note Supplier Ledger	Material Issue Slip Stock Transfer Note Job / Work Order Sheet Production Entry Slip QC Inspection Report Certificate of Analysis Item & Batch Labels Packing List	Salary Slip Attendance Register Machine History Card Trial Balance / P&L / Balance Sheet GSTR-1 & GSTR-3B working Stock Valuation Report

8.3 User stories — printing

US-PDF-01 — As an **Accountant**, I want to print a tax invoice that already matches our old printed format, so that customers do not notice a disruptive change.

Acceptance: the template editor allows adjusting the header block, column set, column widths and footer terms without a developer; the result matches the sample provided during implementation.

US-PDF-02 — As a **Store Keeper**, I want to print a 100×50 mm batch label with a QR code from the GRN screen, so that every incoming lot is identified on the rack.

Acceptance: label size is configurable; the QR contains item code, batch number and GRN number; printing works on a standard thermal label printer through the browser.

US-PDF-03 — As a **Sales Executive**, I want to send a quotation on WhatsApp directly from the quotation screen, so that I do not download, switch apps and re-upload.

Acceptance: one button generates the PDF, stores it, creates a signed link and queues the WhatsApp template message; the screen shows the delivery status once the webhook returns.

9. Customization Framework

The same product will be sold to eleven kinds of factory. Every one of them will ask for something slightly different. If each request becomes a code change, the product becomes unmaintainable. So the following seven things are customisable **as data**, by the customer or the implementer, with no deployment:

#	What is customisable	How it works
C1	Custom fields	Any master or transaction screen can receive extra fields defined in Settings: field label, key, data type (text, number, decimal, date, dropdown, multi-select, checkbox, file), required or optional, default value, validation (min, max, regex, options list), display section, sort order, and whether it appears on the printed document, in list columns and in exports. Values are stored in a JSON column on the parent table plus a normalised <code>custom_field_values</code> table for filterable fields.
C2	Document numbering	Per document type and per financial year: prefix, suffix, padding width, starting number, separator, and whether the number resets each year. Example pattern <code>{PREFIX}/{FY}/{SEQ:5}</code> produces <code>S0/26-27/00042</code> . Numbers are allocated with a row lock so two simultaneous saves cannot collide. Optional per-branch series.
C3	Approval workflows	Rules per document type with conditions on any header field (amount, discount %, customer group, item category), a sequence of approver roles, an optional parallel or sequential mode, escalation after N hours, and auto-approval below a threshold.
C4	Print templates	As described in Section 8. Multiple templates per document type, one marked default, selectable at print time.
C5	Terms and conditions library	Named blocks of text with rich formatting, attachable per document type and per customer, so export customers get different terms from domestic ones.
C6	Dashboards	Each role gets a dashboard assembled from a widget library (KPI tile, line chart, bar chart, pie, table, pending-approvals list). Widgets are chosen, ordered and resized per role; users may then rearrange their own copy.
C7	Labels and language	Every visible label can be overridden — a furniture maker may call a Work Order a "Job Card". Language files support English, Hindi and Gujarati, with the option to add more by uploading a translation file.

Guard rail. Custom fields must never be allowed on columns that drive calculations (quantity, rate, tax, stock). Those belong to the core model. Custom fields are for information the business wants to record and print, not for changing arithmetic. Any request to change a calculation is a code change and goes through the release process.

10. Functional Modules

10.1 How to read a module specification

Every module below follows the same structure, so nothing is missed:

Part	What it tells you
Purpose	Why the module exists, in one or two sentences.
How it works	The step-by-step flow a user experiences, written as plain narrative.
User stories	Numbered requirements in the form "As a <role>, I want <goal>, so that <benefit>", each with acceptance criteria. These become test cases directly.
Fields and validation	Every input field, its data type, whether it is mandatory, and the exact rules that must be enforced on both the client and the server.
Business rules	Conditions that are not tied to a single field — cross-document checks, status transitions, calculations.
Status flow	The states a document can be in and which transitions are legal.
Permissions and alerts	Which permissions guard the module and which notifications it raises.

Validation is always enforced twice. Client-side validation exists only for a good user experience. Server-side validation in a Laravel FormRequest is the real rule and must never be skipped, because the API and imports bypass the browser entirely.

10.2 Module list

Code	Module	Phase	Depends on
M01	Company Setup and Master Data	1	—
M02	User, Role and Permission Management	1	M01
M03	Item Master, UOM, Batch and Serial	1	M01
M04	Bill of Materials and Routing	2	M03
M05	CRM and Lead Management	3	M01, M02
M06	Sales — Quotation to Invoice	2	M01, M03, M08
M07	Purchase and Vendor Management	2	M01, M03, M08
M08	Inventory and Warehouse	1	M03
M09	Production Planning and Execution	2	M03, M04, M08
M10	Quality Control	3	M03, M07, M09
M11	Machine, Mould and Die Maintenance	3	M01, M09
M12	Dispatch, Packing and Logistics	2	M06, M08
M13	Accounts, Finance and GST	3	M06, M07
M14	HR, Attendance and Payroll	4	M01, M02

Code	Module	Phase	Depends on
M15	Reports, Dashboards and Analytics	2-4	all
M16	Settings, Customization and Utilities	1	M01, M02

MODULE M01

Company Setup and Master Data**Purpose**

Holds the information that every other module reads: the company itself, its branches and warehouses, and the shared lists — customers, suppliers, tax rates, units, categories, transporters. If this data is wrong, every document produced afterwards is wrong, so this module is built first and validated hardest.

How it works

After installation, a setup wizard walks the implementer through six steps: company details and GST registration; financial year and numbering; branches and warehouses; tax and UOM masters; industry profile selection; and the first admin users. Each step can be revisited later from Settings. Masters can also be bulk-imported from Excel using a downloadable template, with a validation preview that shows errors row by row before anything is written.

User stories

US-M01-01 — As a **Super Admin**, I want to enter company details, GSTIN, PAN and upload a logo, so that every printed document carries correct legal information.

Acceptance: the logo appears on all print templates; changing the GSTIN after invoices exist requires a confirmation and is logged.

US-M01-02 — As a **Super Admin**, I want to define multiple warehouses and storage bins, so that stock can be located precisely on the shop floor.

Acceptance: warehouses support a parent-child hierarchy (Plant → Store → Rack → Bin) up to four levels; a warehouse with stock cannot be deleted.

US-M01-03 — As an **Accountant**, I want to add a customer with GSTIN, and have the state code, place of supply and default tax type filled in automatically, so that GST is applied correctly without me remembering the rules.

Acceptance: entering a valid 15-character GSTIN populates the state from the first two digits and sets the tax type to IGST when the customer state differs from the company state, CGST+SGST when it matches.

US-M01-04 — As a **Purchase Officer**, I want to record multiple contact persons and addresses per supplier, so that the PO goes to the right person and material goes to the right gate.

Acceptance: unlimited addresses tagged as billing, shipping or factory; exactly one must be marked default per type.

US-M01-05 — As a **Super Admin**, I want to import 2,000 customers from an Excel file, so that go-live does not take weeks of typing.

Acceptance: a downloadable template with the exact columns; a dry-run preview listing every invalid row with the reason; the import is queued and reports "1,942 imported, 58 skipped" with a downloadable error file.

Fields and validation — Company

Field	Type	Req.	Validation rule
Legal Name	Text	Yes	2-150 characters. Letters, digits, space and . , & () - / only.
Trade Name	Text	No	Max 150 characters. Defaults to legal name if blank.
GSTIN	Text	Cond.	Required if "GST registered" is ticked. Exactly 15 characters matching <code>^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}\$</code> . Checksum digit verified. Characters 3-12 must equal the PAN.
PAN	Text	Yes	10 characters, <code>^[A-Z]{5}[0-9]{4}[A-Z]\$</code> . Auto-uppercased.
CIN	Text	No	21 characters if entered.
Registered Address	Textarea	Yes	10-250 characters.
State	Select	Yes	From the state master. Must match the GSTIN state code when GSTIN is present — otherwise show the error "State does not match GSTIN".
PIN Code	Text	Yes	Exactly 6 digits, first digit 1-9.
Phone	Text	Yes	10-15 digits, optional leading +. Spaces and dashes stripped before saving.
Email	Email	Yes	Valid format, max 100 characters, lower-cased on save.
Logo	File	No	PNG or JPG, max 1 MB, minimum 200×200 px, maximum 2000×2000 px. MIME type verified from file content, not from the extension.
Financial Year Start	Select	Yes	Month and day. Default 1 April. Cannot be changed once a transaction exists in the current year.
Base Currency	Select	Yes	Default INR. Locked after the first transaction.
Decimal Places — Amount	Integer	Yes	0-4, default 2.
Decimal Places — Quantity	Integer	Yes	0-4, default 3.

Fields and validation — Customer / Supplier (shared "Party" master)

Field	Type	Req.	Validation rule
Party Type	Multi-select	Yes	Customer, Supplier or both. A party can be both — many factories buy from and sell to the same trader.
Party Code	Text	Auto	Generated by the numbering engine. Unique. Editable only before the first transaction.
Party Name	Text	Yes	2-150 characters. Duplicate name warning (not a hard block) with a link to the existing record.
GSTIN	Text	Cond.	Same rule as company. Must be unique across parties. Required when GST Type = Registered.
GST Type	Select	Yes	Registered / Unregistered / Composition / SEZ / Export / Deemed Export. Drives the tax logic on invoices.
Billing Address	Composite	Yes	Line 1 required, line 2 optional, city required, state required, PIN required (6 digits), country default India.
Shipping Addresses	Repeatable	No	Zero or more. Exactly one must be marked default when at least one exists.

Field	Type	Req.	Validation rule
Contact Persons	Repeatable	Yes (≥1)	Name required; mobile required and 10 digits for India; email optional but valid if entered; designation optional; WhatsApp opt-in checkbox with the consent date recorded automatically.
Credit Limit	Decimal	No	≥ 0, max 12 digits. Zero means unlimited only if "Unlimited credit" is ticked; otherwise zero blocks all credit sales.
Credit Days	Integer	No	0–365. Drives the invoice due date and the overdue reminder.
Payment Terms	Select	No	From the terms master. Prints on the invoice.
Price List	Select	No	From the price list master. Determines the default rate on sales documents.
TDS / TCS Applicable	Checkbox + Select	No	When ticked, a section and rate must be selected.
Assigned Sales Executive	Select	Cond.	Required when the record scope model is in use, because it drives who can see the party.
Bank Details	Composite	No	IFSC must match <code>^[A-Z]{4}0[A-Z0-9]{6}\$</code> ; account number 9–18 digits; the two account-number entries must match.
Status	Select	Yes	Active / Inactive / Blocked. Blocked parties cannot be used on new documents; existing documents are untouched.

Business rules

#	Rule
BR-01	A master record that is referenced by any transaction can never be deleted — only made inactive. The delete button shows the count of blocking references.
BR-02	GSTIN uniqueness is checked across the whole party table, ignoring soft-deleted rows, and the error message names the conflicting party.
BR-03	Changing a party's state after transactions exist raises a warning, because it changes IGST versus CGST/SGST behaviour on future documents. Past documents keep their original tax treatment.
BR-04	Warehouses form a tree; a parent cannot be deleted while children exist, and stock is only ever held at leaf level.
BR-05	The company record is a singleton — exactly one row, created by the installer, never deletable.

MODULE M02

User, Role and Permission Management

Purpose

Creates the people who use the system and controls exactly what each of them may do and see. The detailed model is described in Section 5; this module is the screens that operate it.

How it works

An admin creates a user, links them optionally to an employee record, assigns one or more roles, and sets a data scope. The permission matrix screen shows a grid of modules down the side and actions across the top, with a tick box in every cell and "select all" controls for a row or a column. Changes take effect on the user's next request — the permission cache is cleared immediately for that user.

User stories

US-M02-01 — As a **Super Admin**, I want to create a user and send them their login details on WhatsApp, so that onboarding a new staff member takes one minute.

Acceptance: the created user receives a WhatsApp template message with the login URL, username and a one-time password valid for 48 hours.

US-M02-02 — As a **Super Admin**, I want to see the full permission list of any user in one screen, so that I can answer "why can't Ramesh approve this?" without guessing.

Acceptance: an "Effective permissions" view lists every permission the user holds, showing whether it came from a role or a direct grant, and which role.

US-M02-03 — As a **Super Admin**, I want to restrict a user to one branch, so that a branch manager cannot see another branch's data.

Acceptance: every list query, report, dashboard tile and export respects the branch scope; documents from other branches return 403 on direct access.

US-M02-04 — As a **Store Keeper**, I want a simple, large-button screen on the shop-floor tablet, so that I can record issues quickly with dusty hands.

Acceptance: roles can be flagged as "simplified UI", which loads a reduced navigation with larger controls and only the permitted screens.

Fields and validation — User

Field	Type	Req.	Validation rule
Full Name	Text	Yes	2-100 characters, letters, spaces, dot and apostrophe.
Username	Text	Yes	4-30 characters, lower-case letters, digits, dot and underscore. Unique. Immutable after creation.
Email	Email	Yes	Valid, unique among active users, max 100 characters.
Mobile	Text	Yes	10 digits for India, must not start with 0-5. Unique. Used for WhatsApp alerts.
Password	Password	Yes	Minimum 8 characters with at least one letter and one digit; confirmation must match; must not contain the username; checked against a common-password list.
Roles	Multi-select	Yes	At least one role. Only Super Admin may assign the Super Admin role.
Branch	Select	Yes	The user's home branch. Must exist and be active.
Data Scope	Select	Yes	All / Branch / Team / Own. When Team is chosen, a list of team members becomes required.
Linked Employee	Select	No	Must not already be linked to another user.
Allowed Warehouses	Multi-select	Cond.	Required for store and production roles. Restricts which warehouses appear in dropdowns.
Require 2FA	Checkbox	No	Inherited from the role if the role sets it; can be forced on individually but not off.
Valid From / Valid Until	Date	No	Optional access window for contract staff. Valid Until must be after Valid From. Login is refused outside the window.
Is Active	Toggle	Yes	Default on. Turning it off ends sessions, revokes API tokens and deletes device tokens immediately.

Business rules

#	Rule
BR-06	The system must always have at least one active Super Admin. Deactivating or removing the role from the last one is blocked with a clear message.
BR-07	A user cannot edit their own roles, permissions or data scope, even as Super Admin. This prevents accidental self-lockout and privilege escalation.
BR-08	System roles (Super Admin) cannot be deleted or renamed; their permission set cannot be reduced.
BR-09	Deleting a user is a soft delete. Their name remains visible on documents they created, shown as "Ramesh Patel (inactive)".
BR-10	Any change to roles or permissions writes an activity log entry listing exactly which permissions were added and removed.

MODULE M03

Item Master, UOM, Batch and Serial

Purpose

Defines everything the factory buys, makes, stores and sells. This is the most heavily referenced master in the system, and the place where the industry profile has the biggest visible effect.

How it works

An item belongs to a type — Raw Material, Consumable, Semi-Finished, Finished Goods, Packing Material, Scrap, Spare Part or Service. The type controls which sections of the form appear: a Finished Goods item shows sales price and BOM links; a Raw Material shows purchase settings and reorder levels; a Service item hides all stock settings. The industry profile adds its own attribute block (cavity count for plastic, GSM for packaging, alloy for aluminium) as configured fields. Items can carry batch numbers, serial numbers, both, or neither, and this choice is locked once stock exists.

User stories

US-M03-01 — As a **Store Keeper**, I want each item to have a unique code and a scannable barcode, so that receiving and issuing material is fast and error-free.

Acceptance: the item code is auto-generated from a configurable pattern or typed manually; a Code128 barcode is printable from the item screen; scanning the barcode in any item field selects that item.

US-M03-02 — As a **Purchase Officer**, I want a reorder level and reorder quantity per item per warehouse, so that the system tells me what to buy before we run out.

Acceptance: when available stock falls to or below the reorder level, alert N06 fires and the item appears on the "Items to reorder" report with a suggested quantity.

US-M03-03 — As a **QC Inspector**, I want every chemical batch to carry a manufacturing date and expiry date, so that expired material is never issued to production.

Acceptance: for items with expiry tracking, issue screens exclude expired batches, warn on batches expiring within a configurable window, and default to the earliest expiry (FEFO).

US-M03-04 — As a **Sales Executive**, I want to see the available-to-promise quantity while taking an order, so that I do not promise stock that is already committed.

Acceptance: the item picker shows physical stock, committed to other orders, in production, and free-to-sell, per warehouse.

US-M03-05 — As a **Production Planner**, I want to define alternative raw materials for an item, so that production is not stopped when the preferred grade is unavailable.

Acceptance: an item may list substitute items with a conversion ratio; the material issue screen offers substitutes with a visible warning and records which substitute was used.

Fields and validation — Item

Field	Type	Req.	Validation rule
Item Code	Text	Yes	3–30 characters, upper-case letters, digits, dash and slash. Unique including soft-deleted rows. Immutable after the first transaction.
Item Name	Text	Yes	3–150 characters. Duplicate-name warning within the same category.
Item Type	Select	Yes	One of the eight types. Cannot change from a stock type to Service once stock exists.
Category / Sub-category	Select	Yes	From the category tree, maximum three levels. Sub-category options depend on the category chosen.
Stock UOM	Select	Yes	From the UOM master. Immutable once any stock transaction exists.
Purchase UOM	Select	No	Must have a conversion factor to the stock UOM defined for this item.
Sales UOM	Select	No	Same rule as purchase UOM.
UOM Conversions	Repeatable	Cond.	Required if purchase or sales UOM differs from stock UOM. Factor > 0, up to 6 decimals. Circular conversions are rejected.
HSN / SAC Code	Text	Yes	4, 6 or 8 digits for HSN; 6 digits for SAC. Must exist in the HSN master, which also supplies the default GST rate.
GST Rate	Select	Yes	0, 0.25, 3, 5, 12, 18 or 28 percent. Defaults from HSN; an override requires the <code>item.override_tax</code> permission.
Cess Rate	Decimal	No	0–100, up to 2 decimals.
Tracking Type	Select	Yes	None / Batch / Serial / Batch+Serial. Immutable once stock exists. Defaults from the industry profile.
Shelf Life (days)	Integer	Cond.	Required when expiry tracking is on. 1–3650.
Standard Cost	Decimal	No	≥ 0, 4 decimals. Used for valuation when no purchase history exists.
Selling Price	Decimal	Cond.	Required for Finished Goods. ≥ 0. A warning appears if it is below standard cost.
Minimum Selling Price	Decimal	No	≥ 0. Sales below it require the <code>sales_order.approve_below_min</code> permission.
Reorder Level	Decimal	No	≥ 0, per warehouse. Triggers alert N06.
Reorder Quantity	Decimal	Cond.	Required and > 0 when a reorder level is set.
Minimum / Maximum Stock	Decimal	No	Both ≥ 0; maximum must be greater than minimum when both are set.
Lead Time (days)	Integer	No	0–365. Used by material requirement planning.
Default Warehouse	Select	No	Must be an active leaf warehouse.
Weight per Unit	Decimal	Cond.	Required for steel and aluminium profiles. > 0, 4 decimals. Used for weight-based costing and freight.
Item Image	File	No	JPG or PNG, max 2 MB. Up to 5 images, one marked primary.
Drawing / Specification	File	No	PDF, DWG or image, max 10 MB each, with a revision number and revision date. Older revisions are retained.

Field	Type	Req.	Validation rule
Is Purchasable / Sellable / Manufacturable	Checkbox	Yes	At least one must be ticked. Controls where the item appears in dropdowns.
Industry attributes	Dynamic	Varies	Rendered from the industry profile with its own type, required flag and range validation, exactly as defined in Section 4.3.

Batch and serial rules

#	Rule
BR-11	A batch number is unique per item, not globally. The same batch string may exist for two different items.
BR-12	Batch numbers are generated automatically from a configurable pattern (for example {ITEM}/{YY}{MM}/{SEQ:3}) or entered manually, chosen per item.
BR-13	A serial number is unique across the entire system and can exist in only one warehouse, in one state, at any moment.
BR-14	Issuing a batch-tracked item without selecting a batch is blocked at the server. The batch dropdown shows only batches with a positive balance in the selected warehouse.
BR-15	When expiry tracking is on, the default issue order is FEFO (first expiry, first out). The user may override it only with the <code>inventory.override_fefo</code> permission, and the override is logged with a reason.
BR-16	Batch genealogy is maintained: each produced batch records the input batches consumed, so a finished lot can be traced back to raw material lots and forward to the customers it was sold to.

MODULE M04

Bill of Materials and Routing

Purpose

Records the recipe for making an item — which materials, how much of each, plus the sequence of operations, the machines used and the time taken. The BOM drives material planning, material issue, production costing and scrap accounting.

How it works

A BOM has a header (the item being produced, the output quantity the recipe is written for, a version, and a validity period) and two sets of lines: **components** and **operations**. Components list the input items with quantity per output unit and an expected wastage percentage. Operations list the manufacturing steps in sequence, each with a machine or work centre, setup time, run time per unit and an optional outsourced-vendor flag for job work.

BOMs are versioned. When a recipe changes, a new version is created rather than editing the old one, because past production must remain explainable. Only one version can be "active" for a given item at a given date. Multi-level BOMs are supported: a furniture BOM may include a sub-assembly which itself has a BOM, and the system explodes the tree to any depth, detecting circular references.

User stories

US-M04-01 — As a **Production Planner**, I want to define a BOM with materials and wastage, so that the correct quantity is issued and scrap is expected rather than a surprise.

Acceptance: required quantity = (BOM quantity ÷ BOM output) × order quantity × (1 + wastage %); the calculation is shown on screen before the work order is released.

US-M04-02 — As a **Production Planner**, I want to create a new version of a BOM when the design changes, so that old work orders still show what was actually used.

Acceptance: versions are numbered automatically; existing open work orders keep their original version; new work orders pick the active version by date.

US-M04-03 — As a **Costing user**, I want the system to calculate the standard cost of a finished item from its BOM, so that I know whether the selling price is profitable.

Acceptance: cost = sum of component costs (at the chosen valuation) + operation costs (machine rate × time + labour rate × time) + configurable overhead percentage; a "rolled-up cost" view explodes sub-assemblies.

US-M04-04 — As a **Production Planner**, I want to mark an operation as outsourced, so that the system creates a job-work challan instead of a shop-floor task.

Acceptance: outsourced operations require a vendor and a rate; releasing the work order generates a job-work issue with material sent out, and receipt against it reconciles quantity and scrap.

US-M04-05 — As a **Production Planner**, I want to record by-products and scrap that come out of the process, so that grind and off-cuts return to stock with a value.

Acceptance: a BOM may list output items besides the main product, each with an expected quantity and a cost-allocation percentage; production entry receives them into stock automatically.

Fields and validation — BOM header

Field	Type	Req.	Validation rule
BOM Number	Auto	Yes	From the numbering engine. Unique.
Finished Item	Select	Yes	Must be an item with "Is Manufacturable" ticked. Cannot appear as its own component at any level.
Version	Text	Auto	Incremented automatically. Unique per item.
Output Quantity	Decimal	Yes	> 0, up to 4 decimals. The quantity the recipe produces — often 1, sometimes 100 or 1000 for small parts.
Output UOM	Select	Auto	Taken from the item's stock UOM, shown read-only.
Valid From / Valid To	Date	Yes / No	Valid From is required and cannot overlap another active version of the same item. Valid To, if set, must be after Valid From.
Is Active	Toggle	Yes	Only one active version per item per date range.
Overhead %	Decimal	No	0–100, 2 decimals. Applied to the sum of material and operation costs.
Notes / Process Instructions	Textarea	No	Max 2000 characters. Prints on the work order sheet.

Fields and validation — BOM component line

Field	Type	Req.	Validation rule
Component Item	Select	Yes	Must exist and be active. Cannot equal the finished item. Circular reference across levels is detected and rejected with the path shown.
Quantity	Decimal	Yes	> 0, up to 4 decimals. This is the quantity needed for the whole BOM output quantity, not per unit.
UOM	Select	Yes	Must be convertible to the component's stock UOM.
Wastage %	Decimal	No	0–100, 2 decimals. Default 0. A value above 25% shows a confirmation prompt because it is usually a typing error.

Field	Type	Req.	Validation rule
Is Critical	Checkbox	No	When ticked, a work order cannot be released if this component is short.
Substitute Items	Repeatable	No	Alternative item plus conversion ratio > 0.
Operation Sequence	Select	No	Which operation consumes this component. Blank means issue at the start.
Issue Method	Select	Yes	Manual issue / Backflush. Backflush deducts stock automatically when production output is recorded.

Fields and validation — BOM operation line

Field	Type	Req.	Validation rule
Sequence	Integer	Yes	> 0, unique within the BOM. Determines the order of operations.
Operation	Select	Yes	From the operation master (Mixing, Extrusion, Moulding, Cutting, Printing, Assembly, Anodising, Packing ...).
Work Centre / Machine	Select	Cond.	Required when not outsourced. Must be an active machine.
Setup Time (min)	Decimal	No	≥ 0. Counted once per work order, not per unit.
Run Time per Unit (min)	Decimal	Yes	> 0. For continuous processes such as extrusion, enter time per metre or per Kg to match the stock UOM.
Machine Rate per Hour	Decimal	No	≥ 0. Defaults from the machine master.
Labour Rate per Hour	Decimal	No	≥ 0.
Operators Required	Integer	No	0–20. Multiplies the labour cost.
Is Outsourced	Checkbox	No	When ticked, vendor and rate per unit become required and machine becomes optional.
Quality Check Required	Checkbox	No	When ticked, an in-process inspection is created automatically after this operation, and the next operation is blocked until it passes.

MODULE M05

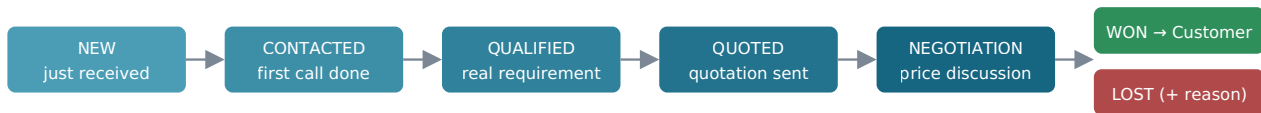
CRM and Lead Management

Purpose

Captures every enquiry from the moment it arrives until it becomes a customer or is lost, so that no enquiry is forgotten and the owner can see where sales effort is going.

How it works

Leads enter the system from four sources: manual entry, a website enquiry form, a bulk CSV or Excel upload, or a marketplace API such as IndiaMART or TradeIndia. Each lead is assigned to a sales executive, either manually or by a round-robin rule. The lead moves through six stages. At every stage the executive records follow-ups with a next-action date; the system reminds them by push notification on the due date. A won lead converts into a Party (customer) and optionally straight into a quotation, carrying its details across without retyping.



A lead may be marked LOST from any stage; a mandatory loss reason is captured (price, quality, delivery time, competitor, no response, other).

Backward movement is allowed and logged. Every stage change records who changed it, when, and the remark.

Leads with no activity for N days (default 7) appear on a "stale leads" report and alert the sales manager.

Figure 10.1 — Lead stage flow.

User stories

US-M05-01 — As a **Sales Executive**, I want a reminder on my phone for today's follow-ups, so that I do not lose an enquiry by forgetting to call.

Acceptance: a push notification each morning listing today's follow-ups with a deep link; overdue follow-ups are highlighted in red on the lead list.

US-M05-02 — As a **Sales Manager**, I want duplicate leads detected automatically, so that two executives do not call the same customer.

Acceptance: on save, the system checks mobile number, email and company name against existing leads and customers, and shows possible matches with an option to merge or continue.

US-M05-03 — As a **Sales Manager**, I want to import leads from an IndiaMART export, so that enquiries reach the team the same day.

Acceptance: a mapping screen matches file columns to system fields, remembers the mapping for next time, validates rows, and reports imported and rejected counts.

US-M05-04 — As an **Owner**, I want a funnel report showing how many leads are in each stage and the conversion percentage per executive, so that I can see who is performing.

Acceptance: a funnel chart with drill-down, filterable by date range, source, executive and product category.

Fields and validation — Lead

Field	Type	Req.	Validation rule
Lead Number	Auto	Yes	From the numbering engine.
Company Name	Text	Yes	2-150 characters.
Contact Person	Text	Yes	2-100 characters.
Mobile	Text	Yes	10 digits (India). Duplicate check across leads and parties.
Email	Email	No	Valid format if entered.
Source	Select	Yes	Website / IndiaMART / TradeIndia / Referral / Exhibition / Cold call / Walk-in / Other. "Other" requires a note.
City / State	Text / Select	Yes	State from the master.
Product Interest	Multi-select	Yes	At least one item or item category.
Estimated Value	Decimal	No	≥ 0 . Used for the pipeline value report.
Expected Closing Date	Date	No	Cannot be in the past on creation.
Assigned To	Select	Yes	An active user holding a sales role. Defaults by the round-robin rule.

Field	Type	Req.	Validation rule
Stage	Select	Yes	Defaults to New. Moving to Won requires conversion; moving to Lost requires a reason.
Priority	Select	No	Hot / Warm / Cold. Default Warm.
Next Follow-up Date	Date	Cond.	Required unless the stage is Won or Lost. Cannot be more than 180 days ahead.
Follow-up Log	Repeatable	No	Date, mode (call, visit, email, WhatsApp), remark (max 500 characters), next action date. Entries cannot be edited after 24 hours, only appended.

MODULE M06

Sales — Quotation to Invoice

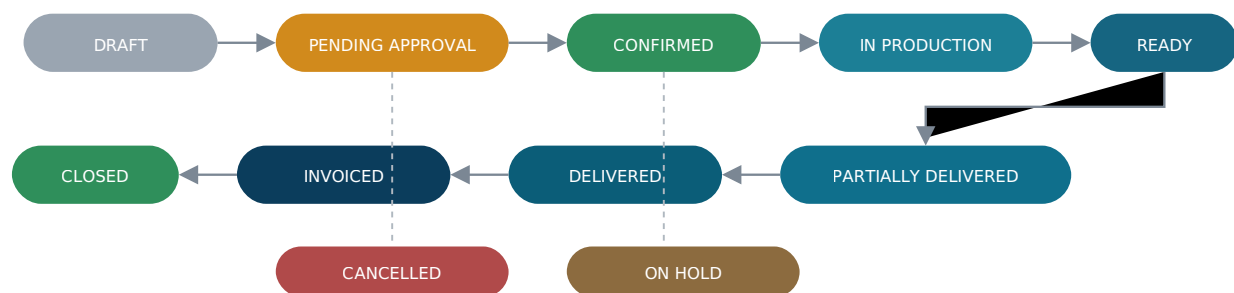
Purpose

Runs the complete order-to-cash cycle: quoting a price, taking a confirmed order, reserving or producing the goods, delivering them and raising a GST-compliant invoice.

How it works

A **quotation** is prepared with items, quantities, rates from the customer's price list, discounts and taxes, and has a validity date. If the customer accepts, one click converts it into a **sales order** without retyping. The sales order checks stock availability and the customer's credit limit. If either fails, it routes to the appropriate approver rather than being blocked outright. A confirmed order **commits** stock — the quantity is no longer free to sell to someone else, though it is still physically present.

From the sales order, the business either dispatches from stock or raises a work order for production. Dispatch creates a **delivery challan** which physically reduces stock. The **tax invoice** can be raised from the challan (common for regular supply) or directly from the order (common when goods and invoice travel together). Partial deliveries and partial invoicing are supported throughout, with each document showing pending quantities.



Legal transitions only. A CONFIRMED order can be cancelled (releasing committed stock) but an INVOICED order cannot — a credit note is required instead.
 ON HOLD can be entered from CONFIRMED or IN PRODUCTION (customer asked to pause) and returns to the same state it left.
 Every transition writes an activity log entry with the old status, new status, user, timestamp and reason where a reason is required.

Figure 10.2 — Sales order status flow.

User stories

US-M06-01 — As a **Sales Executive**, I want the item rate to fill in automatically from the customer's price list, so that I quote consistent prices.

Acceptance: rate is picked in this order — customer-specific rate, then customer group price list, then item selling price. The source of the rate is shown as a tooltip.

US-M06-02 — As a **Sales Executive**, I want to convert an accepted quotation into a sales order in one click, so that nothing is re-typed and nothing is mistyped.

Acceptance: all header and line data carry across; the order links back to the quotation; the quotation status becomes "Converted" and can no longer be edited.

US-M06-03 — As a **Sales Manager**, I want orders that exceed the customer's credit limit to need my approval, so that we do not increase bad debt.

Acceptance: on save, the system computes current outstanding + open orders + this order value; if it exceeds the limit, the status becomes Pending Approval, alert N17 fires, and no dispatch is possible until approved.

US-M06-04 — As a **Sales Executive**, I want to see how much of each ordered item is still pending, so that I can answer the customer's "where is my material" call immediately.

Acceptance: each order line shows ordered, produced, delivered, invoiced and pending quantities, updated in real time.

US-M06-05 — As an **Accountant**, I want GST calculated correctly for interstate and intrastate customers, so that returns match the books.

Acceptance: IGST when the place of supply state differs from the company state, CGST + SGST when it matches, zero-rated for exports and SEZ with the correct LUT flag; the tax breakdown is shown line by line and summarised by rate.

US-M06-06 — As a **Sales Executive**, I want to give a line discount and an overall bill discount, so that I can match how we actually negotiate.

Acceptance: both percentage and absolute discount are supported at line and header level; discount above a configurable percentage triggers approval; the net rate after discount is always shown.

US-M06-07 — As an **Accountant**, I want to raise a credit note against a returned invoice, so that the customer's account and stock are both corrected.

Acceptance: the credit note references the original invoice, cannot exceed its value or quantity, returns the goods to stock at their original cost, and appears in the GST return with the original invoice reference.

Fields and validation — Sales Order header

Field	Type	Req.	Validation rule
Order Number	Auto	Yes	From the numbering engine, allocated on first save with a row lock. Never reused, even if the order is later cancelled.
Order Date	Date	Yes	Cannot be in a closed financial period. Cannot be in the future beyond a configurable tolerance (default 0 days). Cannot be earlier than the linked quotation date.
Customer	Select	Yes	Must be an active party of type Customer. Blocked parties are rejected with the block reason shown.
Customer PO Number	Text	Cond.	Max 50 characters. Required when the customer's "PO mandatory" flag is set. Duplicate PO number for the same customer raises a warning.
Customer PO Date	Date	Cond.	Required with the PO number. Cannot be after the order date.
Quotation Reference	Select	No	Must belong to the same customer and be in Sent or Accepted status.
Billing Address	Select	Yes	Defaults to the customer's default billing address; editable per order.
Shipping Address	Select	Yes	Defaults to the default shipping address. Drives the place of supply and therefore the tax type.

Field	Type	Req.	Validation rule
Place of Supply	Select	Yes	Auto-filled from the shipping address state; editable with the <code>sales_order.override_pos</code> permission.
Expected Delivery Date	Date	Yes	Cannot be before the order date. A warning appears if it is earlier than today plus the longest item lead time.
Warehouse	Select	Yes	Must be within the user's allowed warehouses.
Currency / Exchange Rate	Select / Decimal	Yes	Rate > 0, 6 decimals. Locked to 1 for the base currency.
Payment Terms	Select	Yes	Defaults from the customer.
Freight / Packing / Insurance	Decimal	No	≥ 0 each. Each has a taxable flag and its own HSN when taxable.
Round Off	Decimal	Auto	Calculated to the nearest rupee, within ±0.99. Cannot be edited manually.
Terms & Conditions	Rich text	No	Defaults from the terms library.
Internal Remarks	Textarea	No	Max 1000 characters. Never printed on customer documents.
Attachments	File	No	Up to 10 files, 10 MB each. Allowed types: PDF, JPG, PNG, XLSX, DOCX. Extension and MIME type both checked.

Fields and validation — Sales Order line

Field	Type	Req.	Validation rule
Item	Select	Yes	Must be active with "Is Sellable" ticked. The same item may repeat on multiple lines only if "allow duplicate lines" is enabled; otherwise the quantities must be merged.
Description	Text	Auto	Defaults from the item, editable up to 500 characters — customers often need their own part description.
Quantity	Decimal	Yes	> 0, up to 4 decimals, maximum 9,999,999. Must be a multiple of the item's minimum order quantity if one is set.
UOM	Select	Yes	Must be linked to the item. Changing it recalculates the base quantity.
Rate	Decimal	Yes	≥ 0, 4 decimals. Below the item's minimum selling price requires approval. Zero rate requires the <code>sales_order.allow_free</code> permission and a mandatory reason (sample, replacement, warranty).
Discount % / Amount	Decimal	No	Percentage 0–100 with 2 decimals; amount cannot exceed quantity × rate. Only one of the two can be entered per line.
Taxable Value	Decimal	Auto	quantity × rate – discount. Read-only.
GST Rate	Select	Auto	From the item; override needs permission and is logged.
Tax Amount	Decimal	Auto	Calculated per line, rounded to 2 decimals at line level, then summed — never calculated on the total, because the two methods differ by paise and GST returns are checked line by line.
Line Total	Decimal	Auto	Taxable value + tax + cess.
Expected Delivery Date	Date	No	Per line, for staggered schedules. Defaults to the header date.

Business rules

#	Rule
BR-17	An order must contain at least one line. Saving with zero lines is rejected.
BR-18	Confirming an order creates stock commitments per line and warehouse. Cancelling releases them. Delivering converts commitment into an actual stock reduction.
BR-19	Delivered quantity can never exceed ordered quantity plus a configurable tolerance (default 0%, commonly set to 2% for continuous processes such as extrusion and textiles).
BR-20	Invoiced quantity can never exceed delivered quantity when invoicing is set to follow delivery.
BR-21	Once any dispatch exists against an order, header fields that affect tax (customer, place of supply, currency) become read-only.
BR-22	A confirmed invoice is immutable. Corrections are made only through a credit note or debit note, never by editing the invoice.
BR-23	Invoice numbering must be strictly sequential with no gaps within a financial year, as required by GST rules. Cancellation keeps the number and marks the invoice cancelled.

MODULE M07

Purchase and Vendor Management

Purpose

Controls what the factory buys and from whom, so that material arrives on time, at an agreed price, from an approved supplier, and is checked before it is paid for.

How it works

A department raises a **purchase indent** — a request to buy — which is approved by a manager. Approved indents can be combined into an **RFQ** sent to several suppliers. Quotes received are entered and compared side by side in a **comparative statement** that highlights the lowest price, best delivery time and best payment terms. The buyer selects a supplier, and a **purchase order** is raised, approved according to value rules, and sent to the supplier by WhatsApp or email as a PDF.

When material arrives, the store keeper creates a **goods receipt note** against the PO. The GRN records what actually arrived, which may be less than ordered or of a different quality. If the item requires inspection, the material goes into a quarantine location and stock becomes usable only after QC accepts it. The supplier's **bill** is then matched three ways — PO, GRN and bill — and any mismatch in rate or quantity is flagged before payment.

User stories

US-M07-01 — As a **Store Keeper**, I want the system to suggest what to buy based on reorder levels and open orders, so that I do not have to check every item by hand.

Acceptance: a "Purchase suggestion" screen lists items where free stock is at or below the reorder level, showing current stock, committed quantity, open PO quantity and a suggested order quantity; selected rows generate an indent in one click.

US-M07-02 — As a **Purchase Officer**, I want to compare three supplier quotations on one screen, so that I can justify my choice.

Acceptance: a matrix of items against suppliers showing rate, delivery days, payment terms and total; the lowest rate per item is highlighted; selecting a supplier per item is allowed (split ordering), and the reason for not choosing the lowest is mandatory.

US-M07-03 — As a **Store Keeper**, I want to receive part of a PO today and the rest next week, so that partial deliveries are handled properly.

Acceptance: the GRN screen pre-fills pending quantity per line; receiving less leaves the PO partially received; over-receipt beyond tolerance is blocked.

US-M07-04 — As an **Accountant**, I want the supplier bill checked against the PO rate and the GRN quantity before I pay, so that we do not overpay.

Acceptance: a three-way match screen shows PO rate versus bill rate and GRN quantity versus bill quantity, with differences highlighted; a bill outside tolerance cannot be approved without the `purchase_bill.approve_mismatch` permission and a reason.

US-M07-05 — As a **Purchase Manager**, I want a supplier rating based on on-time delivery, quality rejection and price, so that I can drop poor suppliers.

Acceptance: a monthly score per supplier from three weighted components (weights configurable), shown on the supplier screen with a trend and the underlying data.

Fields and validation — Purchase Order

Field	Type	Req.	Validation rule
PO Number / Date	Auto / Date	Yes	As per the numbering engine. Date cannot be in a closed period.
Supplier	Select	Yes	Active party of type Supplier. Suppliers marked "not approved" are blocked when the approved-vendor policy is on.
Indent / RFQ Reference	Select	Cond.	Required when the setting "PO only against approved indent" is on.
Delivery Warehouse	Select	Yes	Active leaf warehouse within the user's scope.
Expected Delivery Date	Date	Yes	Not before the PO date.
Item	Select	Yes	Active item with "Is Purchasable" ticked.
Quantity	Decimal	Yes	> 0. A warning appears if the quantity exceeds six months of average consumption.
Rate	Decimal	Yes	≥ 0. The last three purchase rates for this item and supplier are shown beside the field. A rate more than a configurable percentage above the last rate requires approval.
GST / Cess	Auto	Yes	Reverse charge flag when the supplier is unregistered; then tax is recorded but not paid to the supplier.
Tolerance %	Decimal	No	0–20. Allowed over-receipt against this line.
Requires Inspection	Checkbox	Auto	Defaults from the item. When on, receipts land in quarantine.

Fields and validation — Goods Receipt Note

Field	Type	Req.	Validation rule
GRN Number / Date	Auto / Date	Yes	Date cannot be before the PO date or in a closed period.
Purchase Order	Select	Cond.	Required unless direct receipt without PO is permitted, which needs the <code>grn.without_po</code> permission.
Supplier Invoice No. / Date	Text / Date	Yes	Unique per supplier per financial year. Date cannot be in the future.

Field	Type	Req.	Validation rule
Vehicle Number	Text	No	Pattern <code>^[A-Z]{2}[0-9]{1,2}[A-Z]{0,3}[0-9]{4}\$</code> when entered.
Received Quantity	Decimal	Yes	> 0. Cannot exceed pending quantity plus the line tolerance.
Accepted / Rejected Quantity	Decimal	Yes	Both ≥ 0 ; their sum must equal the received quantity exactly. Rejected quantity requires a reason.
Batch Number	Text	Cond.	Required for batch-tracked items. Auto-generated or manual per item setting.
Manufacturing / Expiry Date	Date	Cond.	Required when shelf life is set. Expiry must be after manufacturing. Material already expired on receipt is blocked.
Serial Numbers	Repeatable	Cond.	Required for serial-tracked items. The count must exactly equal the accepted quantity. Each serial must be globally unique.
Landed Cost Allocation	Decimal	No	Freight, loading and customs distributed across lines by value, quantity or weight. The chosen basis is recorded and the allocated cost is added to the item's inventory value.
Storage Location / Bin	Select	Cond.	Required when bin management is on.

MODULE M08

Inventory and Warehouse

Purpose

Holds the truth about what material exists, where it is, in which batch, and what it is worth. Every other module either reads from it or writes to it, and nothing writes to it except through the stock ledger.

How it works

Stock is never stored as a number that someone can type over. Instead, every movement writes a row into the **stock ledger** — item, warehouse, batch, quantity in or out, rate, value, source document and timestamp. The current balance is the sum of the ledger, kept in a cached balance table that is updated inside the same database transaction. This design means stock can always be explained: clicking a balance shows every movement that produced it.

Four quantities are tracked separately and must never be confused:

Quantity	Meaning
Physical	What is actually lying in the warehouse right now.
Committed	Physically present but already promised to a confirmed sales order or reserved for a work order.
On order	Not yet arrived, but covered by an approved purchase order or a running work order.
Available (free)	Physical – committed. This is the number a sales executive must see before promising delivery.

Transaction types that move stock

Transaction	Effect	Notes
Goods Receipt Note	IN	From a supplier, at purchase rate plus landed cost
Purchase Return	OUT	At the original receipt rate
Material Issue to Production	OUT	Against a work order, at the valuation rate
Production Receipt	IN	Finished or semi-finished goods at computed cost
By-product / Scrap Receipt	IN	At a configured scrap valuation rate

Transaction	Effect	Notes
Material Return from Production	IN	Unused material sent back to stores
Delivery Challan / Invoice	OUT	At the valuation rate; the sale value is separate
Sales Return	IN	At the original issue cost, not the selling price
Stock Transfer	OUT + IN	Two ledger rows; in-transit stock is visible when transfer takes time
Stock Adjustment	IN or OUT	After physical counting; needs a reason and approval
Job Work Issue / Receipt	OUT / IN	Material at a vendor remains the company's stock, shown in a separate "with vendor" location
Opening Stock	IN	One-time at go-live, locked after the opening date

User stories

US-M08-01 — As a **Store Keeper**, I want to issue material against a work order by scanning a barcode, so that issuing is fast and accurate.

Acceptance: scanning an item barcode adds a line; scanning a batch barcode selects the batch; the BOM-suggested quantity is pre-filled and any difference from it must be explained.

US-M08-02 — As an **Owner**, I want to know the value of stock lying in the factory today, so that I know how much money is tied up.

Acceptance: a stock valuation report as at any past date, by warehouse, category and item, using the configured costing method, tallying with the inventory account in the general ledger.

US-M08-03 — As a **Store Keeper**, I want to do a physical count and post the difference, so that the system matches reality.

Acceptance: a count sheet is generated and optionally printed with a blank quantity column; counted quantities are entered or imported; the variance report shows quantity and value difference per item; posting requires approval above a configurable value and writes an adjustment entry with a reason.

US-M08-04 — As a **QC Inspector**, I want received material to sit in quarantine until I clear it, so that unchecked material never reaches production.

Acceptance: quarantine is a system warehouse; its stock is excluded from available quantity; only a QC acceptance can move it to the main store, and only a QC rejection can move it to the rejection area.

US-M08-05 — As an **Owner**, I want to see slow-moving and dead stock, so that I can stop buying it and clear it out.

Acceptance: a report showing items with no movement in a configurable number of days, with quantity, value and last movement date, exportable.

Fields and validation — Stock Ledger entry (system generated)

Field	Type	Req.	Validation rule
Item	FK	Yes	Must exist and be a stock item.
Warehouse	FK	Yes	Must be a leaf warehouse.
Batch / Serial	FK	Cond.	Required per the item's tracking type.
Posting Date & Time	DateTime	Yes	Cannot be in a closed period. Back-dated entries need the <code>inventory.backdate</code> permission and trigger revaluation of all later entries for that item.

Field	Type	Req.	Validation rule
Quantity In / Out	Decimal	Yes	Exactly one of the two is greater than zero; both cannot be non-zero, and both cannot be zero.
Rate	Decimal	Yes	≥ 0 , 4 decimals. For outward entries, computed by the costing method rather than entered.
Value	Decimal	Auto	quantity $\times$ rate, 2 decimals.
Running Balance Qty / Value	Decimal	Auto	Recomputed in sequence. Balance quantity can never go negative unless negative stock is explicitly allowed in Settings — the default is to block it.
Source Document	Polymorphic	Yes	Type and ID of the document that caused the movement. No ledger row may exist without one.

Business rules

#	Rule
BR-24	Negative stock is blocked by default. When a business genuinely needs it (goods dispatched before the receipt is entered), it is enabled per warehouse, and every negative balance appears on a daily exception report.
BR-25	Stock ledger rows are never updated or deleted. A correction is a reversing entry plus a new entry, both linked to the correction document.
BR-26	All stock writes happen inside a database transaction with a row-level lock on the item-warehouse balance, so two simultaneous issues cannot both consume the last unit.
BR-27	Back-dated entries trigger recalculation of the running balance and valuation for all later entries of that item and warehouse. This runs as a queued job with the item locked during the rebuild.
BR-28	The total value of the stock ledger must always equal the inventory control account in the general ledger. A daily reconciliation job checks this and raises an alert on any difference.

MODULE M09

Production Planning and Execution

Purpose

Turns confirmed demand into a plan, the plan into work orders, and work orders into finished goods — while recording exactly what was consumed, produced, rejected and how long it took.

How it works

The planner starts from demand: confirmed sales orders, forecast quantities, or a make-to-stock decision. A **production plan** collects this demand for a period, explodes the BOM to work out what must be made and what must be bought, compares it with available stock and open purchase orders, and produces two outputs — a list of work orders to create and a list of shortages to purchase.

Each **work order** is for one item and one quantity, links to the BOM version in force, lists the operations to perform, and is assigned to a machine and a target date. Releasing it checks material availability and reserves stock. On the shop floor the operator records **production entries** per shift: quantity produced good, quantity rejected with a defect reason, downtime with a reason, and the operator and machine involved. Materials are either issued manually before production or backflushed automatically when output is recorded, depending on the BOM setting.

Closing a work order computes actual cost — material consumed at valuation rate, machine hours at machine rate, labour hours at labour rate, plus overhead — and compares it against the standard cost from the BOM, producing a variance the owner can act on.

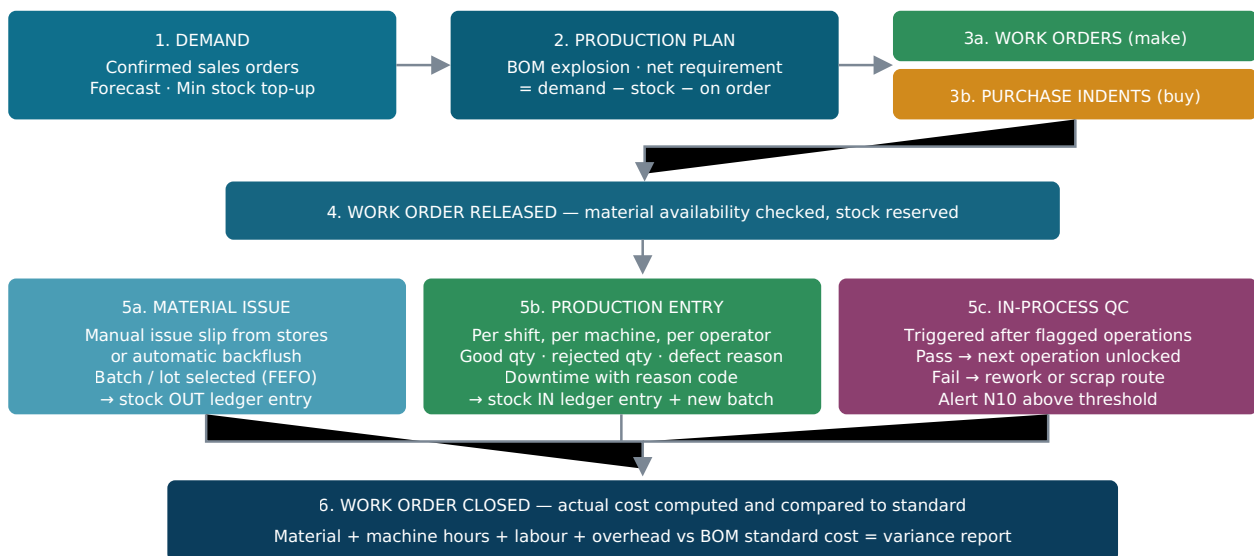


Figure 10.3 — Plan-to-produce flow.

User stories

US-M09-01 — As a **Production Planner**, I want to generate work orders directly from confirmed sales orders, so that nothing ordered is left unmade.

Acceptance: selecting sales order lines creates work orders with the correct item, quantity, BOM version and required-by date; each work order links back to its sales order so progress is visible from the order screen.

US-M09-02 — As a **Production Planner**, I want to see whether material is available before I release a work order, so that the shop floor does not stop halfway.

Acceptance: the release screen shows each component with required, available and shortage quantities; a shortage of a component marked critical blocks release; a shortage of a non-critical component shows a warning and requires confirmation.

US-M09-03 — As a **Shop Floor Operator**, I want to record my shift's output on a tablet in under thirty seconds, so that data entry does not slow down production.

Acceptance: a simplified screen showing only the operator's assigned machine and open work order, with large numeric input for good and rejected quantity and a dropdown of defect reasons; saving is possible offline and syncs when connectivity returns.

US-M09-04 — As a **Production Planner**, I want to see machine-wise loading for the next seven days, so that I do not promise a delivery date the factory cannot meet.

Acceptance: a chart per machine showing planned hours against available hours per day, accounting for shifts, holidays and planned maintenance; overloaded days appear in red.

US-M09-05 — As an **Owner**, I want to know the real cost of every batch produced, so that I know which products actually make money.

Acceptance: a work order costing sheet showing standard versus actual for material, labour, machine and overhead, with the variance in value and percentage, and a drill-down to the transactions behind each figure.

US-M09-06 — As a **Production Planner**, I want rejected output to be routed either to rework or to scrap, so that the loss is recorded honestly.

Acceptance: rejection at production entry requires a defect reason and a disposition (rework, scrap, or downgrade to a second-quality item); rework creates a linked work order, scrap posts a scrap receipt at scrap value, downgrade receives stock against a different item code.

Fields and validation — Work Order

Field	Type	Req.	Validation rule
Work Order Number	Auto	Yes	From the numbering engine.
Item to Produce	Select	Yes	Must have "Is Manufacturable" ticked and an active BOM. Without a BOM the system offers to create one first.
BOM Version	Select	Yes	Defaults to the version active on the work order date. Choosing an older version requires a reason.
Planned Quantity	Decimal	Yes	> 0. A warning appears if it is not a multiple of the item's standard batch size.
Sales Order Reference	Select	No	Must be a confirmed order containing this item. Quantity across all linked work orders cannot exceed the pending order quantity plus tolerance.
Planned Start / End Date	Date	Yes	End cannot be before start. End should not be after the sales order delivery date — a warning appears if it is.
Machine / Work Centre	Select	Cond.	Required when machine scheduling is on. The machine must be active and not under planned maintenance for the whole period.
Shift	Select	No	From the shift master.
Source Warehouse	Select	Yes	Where components are drawn from.
Target Warehouse	Select	Yes	Where finished output is received. Often a work-in-progress or finished-goods store.
Priority	Select	No	Low / Normal / High / Urgent. Drives ordering on the shop-floor list.
Status	Select	Auto	Draft → Released → In Progress → Completed → Closed, with Cancelled and On Hold as exceptions. Backward transitions need the <code>work_order.reopen</code> permission.

Fields and validation — Production Entry

Field	Type	Req.	Validation rule
Work Order	Select	Yes	Must be Released or In Progress. Draft, Completed and Cancelled orders are not selectable.
Entry Date & Shift	Date / Select	Yes	Date cannot be in the future or in a closed period. One entry per work order, machine and shift unless multiple entries are allowed in Settings.
Machine	Select	Yes	Must be active. A machine flagged as under breakdown cannot receive production for that period.
Operator(s)	Multi-select	Yes	Active employees marked as operators. Used for labour costing and productivity reports.
Good Quantity	Decimal	Yes	≥ 0. Cumulative good quantity cannot exceed planned quantity plus the over-production tolerance (default 5%, configurable).
Rejected Quantity	Decimal	No	≥ 0. Requires a defect reason and a disposition when greater than zero.
Defect Reason	Select	Cond.	From the defect master, filtered by the industry profile (short mould, black spot, bubble, off-shade, burr, dimension out, weld defect...).
Rework Quantity	Decimal	No	≥ 0. Cannot exceed rejected quantity.
Batch Number	Text	Cond.	Required for batch-tracked output. Auto-generated by pattern by default.
Start / End Time	Time	Yes	End after start. Total run time cannot exceed the shift duration plus permitted overtime.

Field	Type	Req.	Validation rule
Downtime (minutes)	Integer	No	0 to shift duration. Requires a downtime reason when greater than zero (no material, machine breakdown, power cut, mould change, no operator, planned maintenance).
Consumed Materials	Repeatable	Cond.	Auto-filled from the BOM for the produced quantity when backflush is on; editable with a variance reason if actual consumption differs by more than a configurable percentage.
By-products / Scrap	Repeatable	No	Item, quantity ≥ 0 , warehouse. Received into stock at scrap valuation rate.

Business rules

#	Rule
BR-29	A work order cannot be released if a component marked critical is short. The shortage list is shown with the quantity needed and the earliest expected purchase receipt.
BR-30	Backflush consumption uses the BOM quantity for the good quantity produced, including wastage percentage. If the resulting consumption would drive a component negative, the entry is blocked and the user is told which component is short.
BR-31	Cost of the produced item = total material consumed value + (machine hours $\times$ machine rate) + (labour hours $\times$ labour rate) + overhead percentage, divided by good quantity. Scrap and by-product value is credited before division.
BR-32	Closing a work order is irreversible without the reopen permission and reverses nothing; a wrongly closed order is corrected by a stock adjustment with a reason.
BR-33	Rejection percentage above a configurable threshold (default 5%) triggers alert N10 to QC and management immediately, not at the end of the shift.
BR-34	Batch genealogy is written at production entry: the output batch records every input batch consumed, with quantities, enabling both backward and forward traceability.

MODULE M10

Quality Control

Purpose

Ensures material is checked at three points — when it arrives, while it is being made, and before it is dispatched — and that the results are recorded, traceable and reportable to customers who ask for a certificate.

How it works

A **QC template** is defined per item or item category and per inspection stage. It lists the parameters to test, each with a type (numeric with minimum, maximum and target; a pass/fail choice; or a text observation), a test method reference, and whether the parameter is critical. When a triggering event occurs — a GRN posted, an operation completed, a batch finished — the system creates an **inspection** automatically and notifies QC.

The inspector records readings against a sample. The system evaluates each parameter against its limits and proposes an overall result. A failure of any critical parameter forces rejection. The inspector then records the disposition: accept, accept with deviation (requires an approver and a note), rework, or reject. Stock moves accordingly — from quarantine to the main store on acceptance, or to a rejection area on rejection. For customers who require it, a Certificate of Analysis is generated as a PDF from the recorded readings.

User stories

US-M10-01 — As a **QC Inspector**, I want an inspection created automatically when material is received, so that nothing slips through unchecked.

Acceptance: posting a GRN for an item with "requires inspection" creates a pending inspection, moves stock to quarantine, and pushes alert N05 to QC.

US-M10-02 — As a **QC Inspector**, I want the system to tell me the sample size, so that inspection follows our sampling plan rather than guesswork.

Acceptance: sample size is calculated from the lot quantity using a configurable plan (fixed number, percentage, or square-root-plus-one), and the plan used is shown on the inspection report.

US-M10-03 — As a **Production Planner**, I want the next operation blocked until in-process inspection passes, so that we do not add value to defective material.

Acceptance: operations flagged "QC required" create an inspection on completion; the following operation cannot be started while the inspection is pending or failed.

US-M10-04 — As a **Customer-facing Sales Executive**, I want to send a Certificate of Analysis with the dispatch, so that the customer's incoming inspection is straightforward.

Acceptance: the certificate lists the batch, test parameters, specified limits, actual readings and the pass result, carries the QC signature image, and can be attached to the WhatsApp dispatch alert.

US-M10-05 — As a **QC Head**, I want a Pareto chart of defect reasons for the month, so that we fix the biggest problem first.

Acceptance: defect quantity by reason, sorted descending with a cumulative percentage line, filterable by item, machine, shift, operator and supplier.

Fields and validation — Inspection

Field	Type	Req.	Validation rule
Inspection Number / Date	Auto / Date	Yes	Date cannot be before the source document date.
Inspection Type	Select	Yes	Incoming / In-process / Final / Pre-dispatch / Customer return.
Source Document	Select	Yes	GRN, work order or delivery challan, matching the inspection type.
Item / Batch	Select	Yes	Must exist on the source document.
Lot Quantity	Decimal	Auto	From the source document, read-only.
Sample Size	Decimal	Yes	> 0 and ≤ lot quantity. Suggested by the sampling plan; a manual override requires a reason.
Parameter Readings	Repeatable	Yes	One row per template parameter. Numeric readings must be within the field's own data range; a reading outside the specification limits is flagged but still accepted so the real value is recorded.
Result per Parameter	Auto	Yes	Pass when within minimum and maximum, otherwise Fail. Computed, not typed.
Overall Result	Select	Yes	Proposed by the system: Fail if any critical parameter fails. The inspector may not override a critical failure to Pass.
Accepted / Rejected / Rework Quantity	Decimal	Yes	All ≥ 0; their sum must equal the lot quantity exactly.
Disposition	Select	Yes	Accept / Accept with deviation / Rework / Reject / Return to supplier. "Accept with deviation" requires an approver with the <code>qc.approve_deviation</code> permission and a mandatory justification.

Field	Type	Req.	Validation rule
Inspector	Auto	Yes	The logged-in user; cannot be changed.
Attachments	File	No	Test certificates, photographs of defects. Up to 10 files, 10 MB each.

MODULE M11

Machine, Mould and Die Maintenance

Purpose

Keeps the assets that actually make the product running: machines, moulds, dies and tools. In plastic, rubber, aluminium and steel, the mould or die is often more valuable and more fragile than the machine, and its life must be tracked in shots or cycles, not only in days.

How it works

Every asset has a register entry with its identification, purchase details, capacity and location. Two kinds of maintenance are tracked. **Preventive maintenance** is scheduled by calendar (every 90 days), by running hours, or by cycle count (every 50,000 shots for a mould), whichever comes first; the system raises a task and an alert before it is due. **Breakdown maintenance** is logged when something fails: the time it stopped, the reason, the action taken, spare parts consumed from stock, the time it restarted, and the resulting downtime cost. Cycle counts increment automatically from production entries, so mould life is tracked without extra data entry.

User stories

US-M11-01 — As a **Maintenance Engineer**, I want an alert before a mould reaches its service interval, so that I service it during a planned stop rather than mid-order.

Acceptance: alert N12 fires at a configurable percentage of the interval (default 90%) and again when due; the mould appears on the maintenance-due dashboard.

US-M11-02 — As a **Production Planner**, I want to see which machines are down right now, so that I can reschedule work orders.

Acceptance: a live machine status board showing Running, Idle, Under breakdown or Under maintenance, with the affected work orders listed against each stopped machine.

US-M11-03 — As an **Owner**, I want to know the total downtime and its cost per machine this month, so that I can decide whether to replace a machine.

Acceptance: a report of downtime hours by reason and machine, valued at the machine's hourly rate plus lost production value, with a month-on-month trend.

US-M11-04 — As a **Maintenance Engineer**, I want spare parts issued against a breakdown to come out of stock, so that spares inventory stays accurate.

Acceptance: the breakdown record has a spare-parts section that creates a stock issue against a maintenance cost centre.

Fields and validation — Asset register

Field	Type	Req.	Validation rule
Asset Code	Text	Yes	3–20 characters, unique, immutable after creation.
Asset Type	Select	Yes	Machine / Mould / Die / Tool / Utility. Controls which fields appear.
Name & Make / Model	Text	Yes	Max 100 characters each.

Field	Type	Req.	Validation rule
Serial Number	Text	No	Unique when entered.
Purchase Date & Value	Date / Decimal	No	Date not in the future; value ≥ 0 .
Location / Department	Select	Yes	From the warehouse or department master.
Capacity	Text + Number	No	For example tonnage for an injection machine, or output per hour for an extruder.
Machine Hour Rate	Decimal	Cond.	≥ 0 . Required when the asset is used for costing.
Cavity Count	Integer	Cond.	Required for moulds. 1-128.
Cycle Time (seconds)	Decimal	No	> 0 . Used to compute theoretical output and efficiency.
Life in Cycles / Shots	Integer	Cond.	> 0 for moulds and dies. The running count increments from production entries.
Service Interval	Composite	No	Any combination of days, running hours and cycles. At least one must be set for preventive maintenance to be scheduled.
Status	Select	Yes	Active / Under maintenance / Under breakdown / Retired. Retired assets cannot be selected on new work orders.
Documents	File	No	Manuals, drawings, warranty papers. Up to 20 files, 20 MB each.

MODULE M12

Dispatch, Packing and Logistics**Purpose**

Gets the finished goods out of the gate correctly and legally: packed and labelled, with a delivery challan, an e-way bill where required, and the customer told that their material has left.

How it works

Dispatch begins with a **packing list** built against a sales order: which items, which batches or serials, packed into how many cartons, bundles, bags or pallets, with gross and net weight. Packing generates labels with QR codes so each package can be identified; the packing unit, its code, its label and every scan against it are specified in full in **M17**, and the package count on the challan is derived from scanned units rather than typed. The **delivery challan** is then created from the packing list, reducing stock and recording the vehicle, driver, transporter and LR number. Where the invoice value and distance require it, e-way bill details are captured and the JSON payload for the government portal is prepared. Once the vehicle leaves, alert N13 goes to the customer on WhatsApp with the vehicle number and expected arrival, and proof of delivery can be uploaded when the signed challan returns.

User stories

US-M12-01 — As a **Dispatch Officer**, I want to build a packing list by scanning batch labels, so that the packing list matches what is physically in the truck.

Acceptance: scanning a batch label adds that batch to the current package; the system warns if the batch is not from the linked sales order's reserved stock; package totals update live.

US-M12-02 — As a **Dispatch Officer**, I want e-way bill fields validated before the vehicle leaves, so that the truck is not stopped at a checkpoint.

Acceptance: when the invoice value exceeds the threshold, transport mode, vehicle number, distance and transporter ID become mandatory; vehicle number format is validated; the e-way bill JSON is downloadable and, where the GSP integration is enabled, submitted directly.

US-M12-03 — As a **Customer**, I want a WhatsApp message when my material is dispatched, so that I can arrange to receive it.

Acceptance: alert N13 sends invoice number, vehicle number, transporter, LR number, e-way bill number and expected delivery date, with the invoice PDF attached.

US-M12-04 — As an **Accountant**, I want the challan and the invoice to always agree, so that GST returns reconcile.

Acceptance: an invoice generated from a challan copies quantities exactly and locks them; a difference is only possible through a documented amendment before the invoice is confirmed.

Fields and validation — Delivery Challan

Field	Type	Req.	Validation rule
Challan Number / Date	Auto / Date	Yes	Date cannot be before the sales order date or in a closed period.
Sales Order	Select	Yes	Must be Confirmed or Partially Delivered, and not On Hold or Cancelled.
Dispatch Quantity per line	Decimal	Yes	> 0 and ≤ pending quantity plus tolerance. Available free stock must cover it.
Batch / Serial selection	Select	Cond.	Required for tracked items; only batches with positive balance in the dispatch warehouse are listed; expired batches are excluded.
Transport Mode	Select	Yes	Road / Rail / Air / Ship.
Vehicle Number	Text	Cond.	Required for road transport. Pattern <code>^[A-Z]{2}[0-9]{1,2}[A-Z]{0,3}[0-9]{4}\$</code> .
Transporter / GSTIN	Select / Text	Cond.	Required when an e-way bill is needed. Transporter GSTIN or transporter ID validated for format.
LR / Docket Number & Date	Text / Date	No	Max 30 characters; date not before the challan date.
Distance (km)	Integer	Cond.	1-4000. Required for the e-way bill; determines its validity period.
Driver Name / Mobile	Text	No	Mobile 10 digits when entered.
Number of Packages	Integer	Yes	> 0. Must equal the package count on the linked packing list.
Gross / Net Weight	Decimal	No	Both ≥ 0; gross must be greater than or equal to net.
E-Way Bill Number	Text	Cond.	12 digits. Required before the vehicle leaves when the value exceeds the threshold.
Proof of Delivery	File	No	Image or PDF, max 5 MB. Uploading it sets the challan to Delivered and stamps the delivery date.

MODULE M13

Accounts, Finance and GST

Purpose

Records the money side of every transaction automatically, so that the books are always in step with the stock and the sales — and produces the statutory outputs Indian manufacturers must file.

How it works

The system uses double-entry accounting with a configurable chart of accounts. Documents created in other modules post their accounting entries automatically through a **posting rule map**: a sales invoice debits the customer and credits sales and GST output; a GRN debits inventory and credits goods-received-not-invoiced; a supplier bill clears that liability and credits the supplier. Users never type these entries — they are generated, which is what keeps stock value and the general ledger in agreement.

On top of the automatic postings, users record **payments, receipts, journal entries, contra entries and expenses**. Receipts are allocated against specific invoices, so the outstanding report is by invoice and by age, not just a customer total. GST output is prepared as working sheets for GSTR-1 and GSTR-3B, reconciled against GSTR-2B data imported from the portal, and exported in the required format.

Automatic posting map

Document	Debit	Credit
Goods Receipt Note	Inventory (by item group)	Goods Received Not Invoiced
Supplier Bill	Goods Received Not Invoiced, GST Input	Supplier (Accounts Payable)
Payment to Supplier	Supplier	Bank / Cash
Material Issue to Production	Work in Progress	Inventory
Production Receipt	Inventory (Finished Goods)	Work in Progress
Scrap Receipt	Inventory (Scrap)	Work in Progress
Delivery / Invoice	Cost of Goods Sold, Customer	Inventory, Sales, GST Output
Receipt from Customer	Bank / Cash	Customer (Accounts Receivable)
Credit Note	Sales Return, GST Output	Customer
Stock Adjustment	Inventory or Stock Variance	Stock Variance or Inventory
Payroll	Salary & Wages, Employer contributions	Salary Payable, Statutory Payables
Depreciation (period end)	Depreciation Expense	Accumulated Depreciation

User stories

US-M13-01 — As an **Accountant**, I want invoices to post to the ledger automatically, so that I do not enter the same transaction twice.

Acceptance: confirming an invoice creates a balanced journal entry visible from the invoice screen; deleting is impossible and reversal is only through a credit note.

US-M13-02 — As an **Accountant**, I want to allocate a single receipt of ₹5,00,000 across four invoices, so that ageing is correct.

Acceptance: the receipt screen lists open invoices oldest first with an auto-allocate button; the allocated total cannot exceed the receipt amount; any unallocated balance stays as an advance against the customer.

US-M13-03 — As an **Owner**, I want an ageing report showing who owes us money and for how long, so that I can chase collections.

Acceptance: buckets of 0-30, 31-60, 61-90 and over 90 days, by customer and by invoice, with a total and a drill-down, exportable and sendable on WhatsApp to the sales team.

US-M13-04 — As an **Accountant**, I want a GSTR-1 working sheet from the system, so that filing takes an hour instead of three days.

Acceptance: B2B, B2C large, B2C small, credit and debit notes, exports and HSN summary sections are produced from confirmed invoices; the totals reconcile to the sales ledger; a JSON or Excel export in the portal's format is available.

US-M13-05 — As an **Owner**, I want a profit and loss statement for any date range, so that I can see how the business is doing without waiting for the year end.

Acceptance: P&L, balance sheet and trial balance for any period, with comparison to the previous period, and drill-down from any figure to the underlying vouchers.

US-M13-06 — As an **Accountant**, I want to lock a period once returns are filed, so that nobody changes a past month by accident.

Acceptance: period locking by date with a per-module override permission; any attempt to post into a locked period is rejected with a clear message naming the lock date.

Fields and validation — Payment / Receipt voucher

Field	Type	Req.	Validation rule
Voucher Number / Date	Auto / Date	Yes	Date not in a closed period and not in the future.
Voucher Type	Select	Yes	Receipt / Payment / Contra / Journal.
Party	Select	Cond.	Required for receipt and payment; not applicable to contra.
Amount	Decimal	Yes	> 0, maximum 12 digits with 2 decimals.
Mode	Select	Yes	Cash / Cheque / NEFT / RTGS / UPI / Card. Cash above the statutory limit raises a warning.
Bank Account	Select	Cond.	Required for all non-cash modes.
Instrument Number & Date	Text / Date	Cond.	Required for cheque. Cheque date cannot be more than 3 months old.
Allocation lines	Repeatable	Cond.	Invoice reference plus allocated amount. Sum of allocations ≤ voucher amount. Allocation per invoice ≤ that invoice's outstanding.
TDS Deducted	Decimal	No	≥ 0. When entered, section and rate become required and a TDS payable entry is posted.
Narration	Text	Yes	5-250 characters. Prints on the voucher and the ledger.

Business rules

#	Rule
BR-35	Every journal entry must balance to zero to the paisa before it is written. An unbalanced entry is impossible, not merely warned about.
BR-36	Confirmed vouchers are never edited or deleted. Corrections are reversal entries that reference the original.
BR-37	Invoice numbering must be gapless and sequential per financial year and per series, as required by GST rules.

#	Rule
BR-38	Tax is computed line by line and rounded at line level, then summed. Never compute tax on the invoice total — the results differ and returns are matched line by line.
BR-39	Bank reconciliation marks entries as reconciled with a bank date; reconciled entries cannot be modified.
BR-40	Year-end closing posts profit or loss to the capital or reserves account, carries balances forward and locks the year. Reopening a closed year requires Super Admin and is prominently logged.

MODULE M14

HR, Attendance and Payroll**Purpose**

Manages the workforce — who is employed, when they worked, what they are owed — with the shift, overtime and piece-rate patterns that factories actually use.

How it works

Employees are recorded with their department, designation, shift, salary structure and statutory details. Attendance arrives three ways: from a biometric device file import, from a mobile punch with location, or by manual entry for staff without devices. The system converts raw punches into attendance days using shift rules — grace period, half-day threshold, late marks, overtime beyond shift hours, and weekly offs and holidays from a calendar.

Payroll processing takes attendance plus the salary structure and computes gross pay, statutory deductions (PF, ESI, professional tax, TDS), advances and loan instalments, and net pay. Piece-rate and production-linked incentives are supported by reading production entries for the operator. Salary slips are generated as PDFs and can be delivered on WhatsApp.

User stories

US-M14-01 — As an **HR Officer**, I want to import the biometric punch file every morning, so that attendance is captured without typing.

Acceptance: the import accepts the device's CSV or Excel format, matches employees by device code, reports unmatched punches, and never creates duplicate records for the same employee, date and punch time.

US-M14-02 — As an **HR Officer**, I want overtime calculated automatically from shift rules, so that wage disputes are avoided.

Acceptance: hours beyond shift duration, above a configurable minimum threshold, are marked as overtime and multiplied by the configured rate; the calculation is visible per employee per day.

US-M14-03 — As a **Shop Floor Supervisor**, I want to pay some operators on a piece-rate basis, so that pay matches output.

Acceptance: a salary structure component of type "piece rate" reads the good quantity produced by that operator from production entries for the period, multiplied by the item's piece rate.

US-M14-04 — As an **Employee**, I want my salary slip on WhatsApp, so that I do not have to ask the office for a copy.

Acceptance: alert N19 sends a password-protected PDF slip after payroll is approved; the password is the employee's date of birth in a configured format.

Fields and validation — Employee (selected fields)

Field	Type	Req.	Validation rule
Employee Code	Auto / Text	Yes	Unique, immutable after creation.
Name	Text	Yes	2–100 characters.
Date of Joining	Date	Yes	Not in the future. Attendance before this date is rejected.
Date of Birth	Date	Yes	Age must be at least 18 years at the joining date — a hard block, since employing minors is illegal.
Department / Designation	Select	Yes	From the masters.
Employment Type	Select	Yes	Permanent / Contract / Temporary / Apprentice.
Shift	Select	Yes	From the shift master; can be overridden per day by a roster.
Salary Structure	Select	Yes	Components must sum correctly; a percentage-of-basic component requires basic to be defined first.
PF / ESI / UAN numbers	Text	Cond.	Required when the corresponding scheme is applicable. UAN is 12 digits; ESI is 17 digits.
Bank account & IFSC	Text	Cond.	Required for bank-transfer salary. IFSC pattern validated; account number entered twice.
Biometric Device Code	Text	No	Unique. Used to match punch imports.
Date of Leaving	Date	No	Must be after joining. Setting it deactivates the employee and blocks future attendance.

Privacy note. Employee personal data is sensitive. Access to salary and personal identification fields must be restricted through field permissions to HR roles only, and every view of a salary record is recorded in the access log.

MODULE M15

Reports, Dashboards and Analytics

Purpose

Turns the data captured everywhere else into decisions. A report that nobody opens is wasted work, so the report set is deliberately focused on the questions factory owners actually ask.

Standard report set

Area	Reports
Sales	Sales register · Item-wise sales · Customer-wise sales · Sales executive performance · Quotation conversion ratio · Pending order status · Order versus dispatch versus invoice reconciliation · Product profitability · Customer ageing
Purchase	Purchase register · Item-wise purchase and rate trend · Supplier-wise purchase · Pending PO status · Purchase versus indent analysis · Supplier performance scorecard · GRN pending inspection
Inventory	Stock summary and stock ledger · Batch-wise and expiry-wise stock · Stock valuation as at a date · Slow-moving and dead stock · Reorder level report · Stock ageing · Physical count variance · Warehouse-wise and bin-wise stock · Material consumption analysis
Production	Daily production report · Machine-wise output and utilisation · Shift-wise efficiency · Operator productivity · Work order status · Planned versus actual quantity and cost · Scrap and rejection analysis · Downtime analysis by reason · Overall equipment effectiveness

Area	Reports
Quality	Inspection register · Rejection Pareto by reason · Supplier quality rating · Batch traceability (forward and backward) · Certificate of Analysis register · Customer complaint analysis
Finance	Day book · Cash and bank book · Trial balance · Profit and loss · Balance sheet · Outstanding receivable and payable ageing · GSTR-1 and GSTR-3B working · GSTR-2B reconciliation · TDS summary · Expense analysis by cost centre
HR	Attendance register · Late and absent summary · Overtime summary · Salary register · Statutory contribution statements · Leave balance

Requirements common to every report

#	Requirement	Detail
R1	Consistent filters	Date range, branch, warehouse, item category, party group and status filters behave identically in every report. Filter selections are remembered per user.
R2	Drill-down everywhere	Any total can be clicked to reveal the transactions behind it, and any transaction opens its source document.
R3	Export	Excel, CSV and PDF from every report. Exports respect field permissions — a user who cannot see rates does not get a rate column in the file.
R4	Scheduling	Any report can be scheduled daily, weekly or monthly and delivered by email or WhatsApp to a list of recipients.
R5	Performance	Reports covering up to one year of data must render within 5 seconds. Heavier reports run as queued jobs and notify the user when ready.
R6	Scope-aware	Every report obeys the user's data scope. A branch user's "total sales" is their branch's total, and the report header states the scope applied.

Dashboard widgets

Role	Default widgets
Management	Today's and month-to-date sales value · Production quantity versus plan · Collection versus target · Top 10 outstanding customers · Stock value trend · Rejection percentage trend · Pending approvals
Sales	My pending orders · Today's follow-ups · My month-to-date sales versus target · Quotations awaiting response · Customer credit alerts
Purchase	Items below reorder level · Pending indents to convert · Overdue purchase orders · GRNs awaiting inspection · Price increase alerts
Production	Work orders due today · Machine loading for the week · Yesterday's output versus plan · Material shortages blocking release · Machines under breakdown
Stores	Pending receipts · Pending issue requests · Items expiring in 30 days · Negative stock exceptions · Today's dispatch list

MODULE M16

Settings, Customization and Utilities

Purpose

The control room. Everything that makes one installation different from another lives here, so that an implementer can fit the product to a customer without a developer.

Settings groups

Group	Contents
Company	Company details, branches, financial year, currency, decimal places, logo, letterhead mode
Industry	Active industry profile, feature flags, item attribute definitions, default UOM set, costing method
Numbering	Series per document type per year and per branch, prefix, padding, reset behaviour, preview of the next number
Custom fields	Field builder per screen with type, validation, placement, print and export flags
Approvals	Rules per document type with conditions, approver roles, sequence and escalation
Print templates	Template editor with live preview, paper size, margins, letterhead handling, copies
Notifications	Alert catalogue with per-alert channel and recipient configuration, quiet hours, digest settings, delivery log
Integrations	WhatsApp Cloud API, Firebase, SMTP, GST portal or GSP hooks, biometric device format, marketplace lead import
Users & roles	Users, roles, permission matrix, data scopes, field permissions, session and password policy
Masters	UOM, categories, HSN, tax rates, states, terms library, defect reasons, downtime reasons, departments, shifts, holidays
Localisation	Language, date format, number format (Indian or international grouping), time zone, label overrides
System	Backup schedule and manual backup, restore, update check, licence details, activity log viewer, queue and job monitor, health check, cache clear

Utilities

Utility	Behaviour
Data import	Downloadable template per master, column mapping that is remembered, dry-run validation with a row-by-row error report, queued execution, and a summary with a downloadable error file.
Backup and restore	Database dump plus the uploads folder, compressed and optionally encrypted. Manual button and scheduled cron. Retention count configurable. Restore requires typing the installation name to confirm.
Update manager	Checks for a new version, shows the changelog, takes an automatic backup, puts the site into maintenance mode, applies files and migrations, clears caches, and rolls back on failure.
Health check	Verifies PHP version and extensions, folder permissions, database connectivity, queue worker heartbeat, cron last-run time, disk space, and integration credentials. Shown as a traffic-light panel to Super Admin.
Recycle bin	Lists soft-deleted records for 90 days with the ability to restore, subject to permission and to referential checks.
Bulk operations	Bulk price update, bulk item status change, bulk reassignment of a sales executive's customers — each producing a single summarised activity log entry plus per-record detail.

MODULE M17

Packing Units, QR Codes and Scan Operations

Purpose

Gives every physical package that leaves the production line — a box, a carton, a drum, a bag, a bundle, a roll, a crate — its own permanent identity, printed as a QR code on a sticker. From that point on the package can be received into stock, moved, picked, loaded and dispatched by scanning it, instead of by typing item codes and quantities into a screen. The same scan is also the fastest way to answer the three questions people ask on the shop floor and at the gate: **what is inside this?**, **whose order is it for?**, and **where should it be?**

How it works

When goods finish production and pass inspection, the operator packs them and creates a **packing unit** in the system — one record per physical package. The system allocates a permanent code, prints a label with a QR code, and the operator sticks it on the box. Packing units can be put inside other packing units: twenty cartons on one pallet, four drums in one crate. This is called **nesting**, and it is what makes a warehouse manageable — one scan of the pallet label handles everything stacked on it.

After that, the QR code is simply an input method. Scanning a label at the goods-in point posts a stock-in. Scanning it at the loading bay against a delivery challan posts a stock-out. Scanning it anywhere else opens a read-only information screen. The scanner can be a phone camera through the browser, or a cheap USB or Bluetooth 2D scanner that behaves like a keyboard — the system supports both, because most factories will buy the ₹3,000 scanner for the gate and use phones everywhere else.

The single most important rule in this module. A QR code does not move stock. It *identifies* what to move; the stock ledger described in M08 still performs the movement and remains the only source of truth for quantity. A packing unit is a container reference recorded *on* a ledger entry, never a second parallel stock system. If the packing unit records and the stock ledger can ever disagree, the design is wrong.

10.17.1 Packing hierarchy

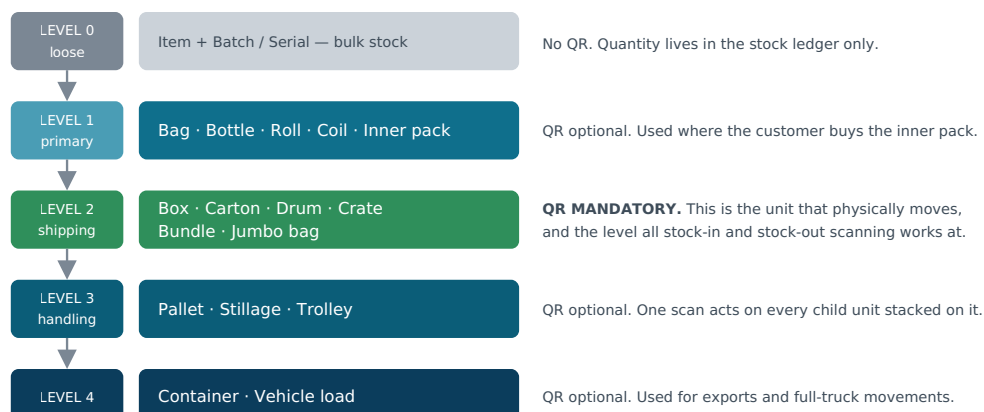


Figure 10.4 — Packing hierarchy. A unit at any level may contain items, or child units, or both. Depth is limited to five levels to keep queries fast on shared hosting.

Which levels a customer actually uses is an **industry profile setting** (§4), not a code change. A PVC pipe plant works in bundles and needs Level 2 only. A chemical plant works in drums with an inner liner and uses Levels 1 and 2. A packaging or auto-parts supplier feeding an OEM will use Levels 2 and 3, because the customer's own goods-in process expects a pallet label.

10.17.2 What the QR code actually contains

This is the decision that is most often made badly, and it is expensive to reverse once ten thousand labels are stuck on drums. The QR code stores **a short random code and nothing else**, wrapped in a URL:

<https://erp.company-domain.com/q/7K4M9XQ2PB>

The code is 10 characters drawn from Crockford Base32 — digits and uppercase letters with I, L, O and U removed because people and OCR confuse them with 1, 0 and V. That gives roughly 10^{15} possible codes, allocated randomly rather than in sequence, with the last character acting as a check digit so a mistyped fallback code is rejected instead of silently matching another package.

Design choice	Reason
Store an identifier, not the data	Printed data goes stale the moment the box moves warehouse, is re-allocated to a different order, or has its batch re-graded after re-testing. An identifier resolved live against the database is never stale.
Random, not sequential	A sequential code tells anyone who scans one label how many units you have shipped this year. It also lets a stranger guess valid codes.
Short payload	Fewer characters means larger, thicker modules at the same printed size, which is what makes a scan work on a dusty drum in poor light. This label needs to survive a warehouse, not a boardroom.
URL form	A phone's default camera app opens it without any app installed — useful for a customer's QC person or a transporter. Staff scanning inside the ERP get the same code without leaving the app.
Never encode price or customer	The label is visible to the transporter, the loading crew and the receiving party. Commercial terms must be looked up behind a login, not read off the box.

Common mistake to avoid. Do not encode a JSON blob of item, batch, quantity, order and customer into the QR. It produces a dense code that fails to scan at arm's length, it leaks pricing and customer names to everyone who handles the box, and it becomes wrong the first time anything about the package changes. Every serious warehouse system stores a licence-plate identifier and looks the rest up.

10.17.3 Label design and physical durability

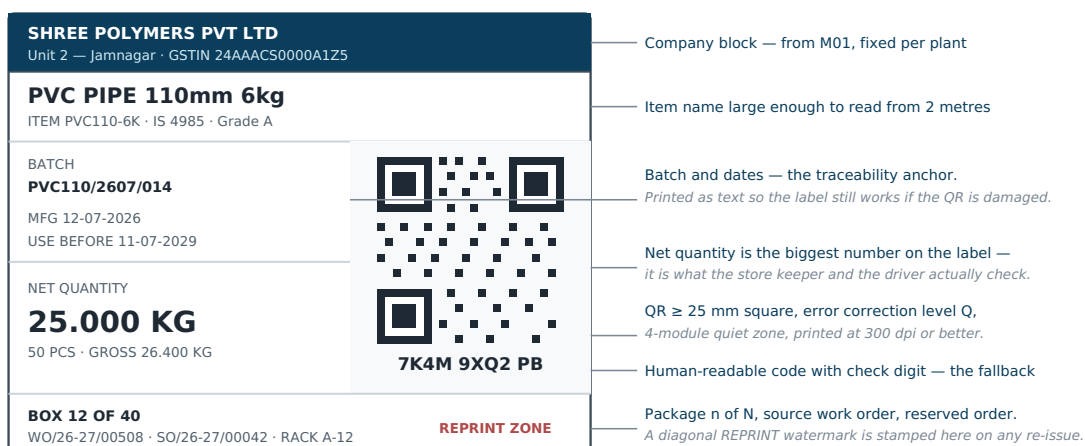


Figure 10.5 — Label anatomy at 100 × 75 mm. Layout, size and which blocks appear are a label template (C4, §9) — not hard-coded.

Package type	Label stock	Why
Carton, box, bundle	Coated paper, permanent adhesive	Cheapest option and adequate. Dry, indoor, short life.
Drum, jerrycan (chemicals)	Polyester or polypropylene synthetic	Survives oil, solvent splash, condensation and outdoor storage. Paper labels lift off drums within days.

Package type	Label stock	Why
Rubber, hot-packed goods	Synthetic with high-temperature adhesive	Applied while the product is still warm; normal adhesive releases.
Steel and aluminium sections	Tag-and-wire, or a large label on the bundle strap	Cannot be stuck to an oiled or scaled surface reliably.
Textile rolls, cable coils	Paper label plus a duplicate inside the wrapper	The outer label is often torn during handling; the inner copy saves the batch identity.
Pallet	A4 sheet in a plastic pouch	Reusable pouch, larger QR, replaced when the pallet is rebuilt.

Printing runs through the existing PDF engine (§8). Labels are generated as a PDF at the exact physical dimensions of the label roll, sent to a thermal label printer. Two practical constraints on shared hosting: the QR image must be produced by a **pure-PHP library** that does not depend on Imagick, and any print run of more than fifty labels must go through the queue rather than the web request, because the PHP execution time limit on cPanel will otherwise cut the job off halfway.

10.17.4 Scanning — one code, several actions

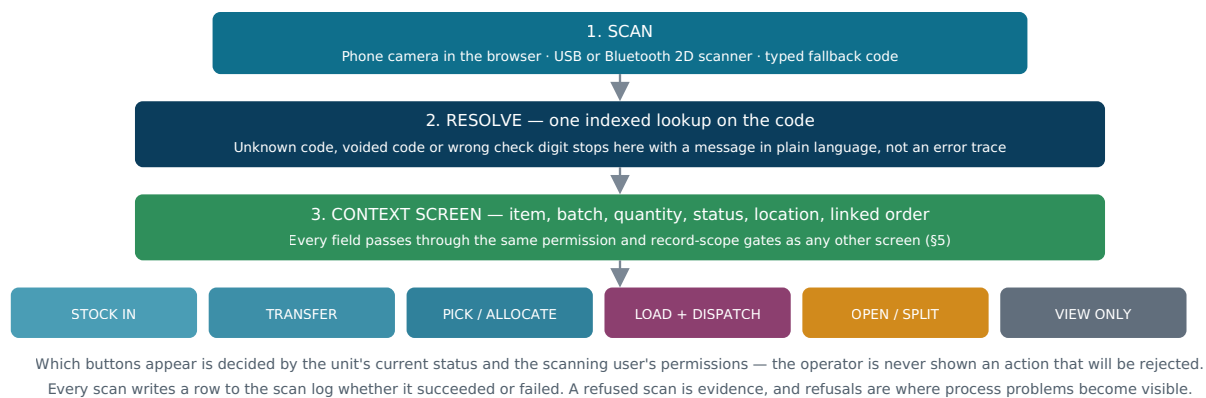


Figure 10.6 — From scan to action. The same code drives receiving, moving, picking, dispatching and enquiry.

Stock in

Three situations produce a stock-in scan. **From production:** after the batch passes QC (M10), the operator packs it, the system creates the packing units, the labels print, and scanning them at the finished-goods store posts the receipt into stock. **From purchase:** incoming material is labelled at the goods-in point after GRN, so raw material drums are traceable through production the same way. **From a customer return:** scanning a returned box brings it back into a quarantine location pending re-inspection, never straight into saleable stock.

Stock out and dispatch

The dispatch officer opens the delivery challan on a phone or tablet and scans each package as it goes onto the vehicle. The screen shows a live count of loaded against required, per item. The value of this is not the speed — it is that the wrong box gets caught at the tailgate instead of at the customer's gate. These are the checks the system runs on every loading scan:

Condition found	Result	What the operator sees
Code not recognised	Block	"This label is not in the system. Do not load this package."
Batch is on QC hold or rejected	Hard block	Names the inspection and its status. No override exists for this, at any permission level.
Batch expired, or remaining shelf life below the customer's agreed minimum	Hard block	Shows expiry date and the customer's shelf-life requirement from the sales order.

Condition found	Result	What the operator sees
Unit already dispatched	Block	Names the challan, date and customer it went out on. This catches a duplicated label immediately.
Unit reserved for a different sales order	Block, overridable	Names the other order and customer. Override needs the reallocation permission and a typed reason, and notifies the sales owner of that order.
Item or batch not on this challan at all	Block, overridable	Offers to add the line if the sales order has a pending quantity for that item; otherwise refuses.
Unit is in a different warehouse than the challan's dispatch warehouse	Warn	Allows loading and posts the implied transfer, so the ledger stays honest instead of the box vanishing.
Unit already scanned in this session	Warn	"Already loaded — package 12 of 40." No second entry is created.
Unit's parent pallet is allocated elsewhere	Block	Prompts the operator to de-nest the unit from the pallet first, so the pallet record does not silently become wrong.
Loaded quantity does not match the challan at confirmation	Block	Lists the shortfall or excess by item and batch. The challan cannot be confirmed until it matches or is amended.

Opening and splitting

Real dispatches are not always whole boxes. Two actions handle this, and both are deliberate rather than silent. **Open** breaks the seal: the unit's status becomes Opened, its contents return to loose stock in the same location, and the code is retired. **Split** takes part of the contents into a new unit with a new code, reducing the original — used when half a drum goes to one customer. Neither action changes total stock quantity. Both are logged with the user, the reason and the before-and-after contents, because a box that quietly loses ten kilograms is how inventory drifts.

10.17.5 What a scan reveals — and to whom

The information screen is a single scan away, which is exactly why it must be filtered. The label is on a box handled by a transporter and opened by a customer.

Information group	Visible to	Notes
Item, batch, quantity, pack date, expiry	Any logged-in user with scan permission	The basic identity of the package.
Current status and storage location	Store, production, dispatch and QC roles	Restricted to the user's branch and warehouse scope.
Source work order, machine, mould, operator, shift	Production, QC, management	The backward traceability chain from §12.3, now anchored at package level.
Inspection results and certificate of analysis	QC, management, and the customer view where enabled	The COA PDF is generated on demand from stored inspection readings.
Linked sales order, customer, challan, invoice	Sales, dispatch, accounts, management	Sales executives see only their own customers unless their data scope is wider.
Cost, selling rate, margin	Management and accounts only	Governed by the field-level rules in §5. Never printed on the label.
Full movement history of this package	Management, audit	Every scan, transfer, allocation and status change with user and timestamp.

Optional customer-facing scan. When the feature flag is on, a person with no login who scans the QR reaches a public page showing item, batch, manufacturing date, expiry, applicable standard and a downloadable certificate of analysis — nothing more. No quantity in stock, no price, no other customer, no order. This is the single most appreciated feature for anyone supplying an OEM or a pharma-adjacent buyer, because their incoming-inspection team can verify a drum in seconds. The page must be rate-limited per IP address and must never accept a query that is not an exact code match.

10.17.6 Working without a network

Godowns and loading bays are where the signal is worst, and that is exactly where scanning happens. The scanning screens therefore operate offline: scans are held in local storage on the device with their timestamp and queued for upload, and the operator continues working with a visible unsent counter. When the connection returns, the queue is submitted in order.

Scans that fail on upload — the package was dispatched by someone else in the meantime, or the challan was cancelled — are never discarded. They land in an **exception list** that the supervisor must clear, with the original scan time preserved so it is clear what happened in what order. A scan queue that silently drops entries is worse than no offline mode at all, because the operator believes the work was recorded.

User stories

US-M17-01 — As a **Packing Operator**, I want to pack a finished batch into boxes and print all the labels in one action, so that forty boxes do not need forty screens.

Acceptance: entering the batch, the quantity per box and the number of boxes creates that many packing units, numbers them "n of N", and produces a single print-ready PDF; the last box automatically carries the remainder when the batch does not divide evenly.

US-M17-02 — As a **Store Keeper**, I want to scan a box to take it into stock, so that I do not type item codes and quantities.

Acceptance: a successful scan posts a stock ledger entry for the exact item, batch and quantity held in the unit, sets the unit's location and status to In Stock, and shows a confirmation with a running count for the session.

US-M17-03 — As a **Dispatch Officer**, I want to load a vehicle by scanning boxes against a challan, so that the wrong material never leaves the gate.

Acceptance: every check in the loading validation table is applied on each scan; the screen shows loaded against required per item; the challan cannot be confirmed while the counts differ.

US-M17-04 — As a **Store Keeper**, I want one scan of a pallet to move all twenty cartons on it, so that a stock transfer takes one action.

Acceptance: scanning a parent unit applies the chosen action to it and every descendant; a preview lists exactly what will be affected before it is committed; any child that fails its own validation blocks the whole action and is named.

US-M17-05 — As a **Quality Inspector**, I want a scanned box to show me who made it, on which machine and mould, and what the test readings were, so that I can investigate a complaint from the box the customer sent back.

Acceptance: the scan screen shows work order, machine, mould, operator, shift, date, time and the full inspection record with readings against specification, in one view without further navigation.

US-M17-06 — As a **Store Keeper**, I want to keep scanning when the warehouse WiFi drops, so that stock taking is not abandoned halfway.

Acceptance: scans queue on the device with a visible unsent count, upload automatically on reconnection, and any that fail appear in a supervisor exception list with their original timestamp — never discarded.

US-M17-07 — As a **Super Admin**, I want reprinted labels controlled, so that two boxes never carry the same code.

Acceptance: a reprint reuses the same code, never generates a new one, requires a reason, prints a visible REPRINT mark, and is logged; a report lists units reprinted more than twice.

US-M17-08 — As a **Store Keeper**, I want to replace a label that has been torn off a drum, so that the drum does not become anonymous stock.

Acceptance: the replace action voids the old code, issues a new one, links old to new permanently, and a later scan of the old code shows "this label was replaced" with the new code rather than a blank error.

US-M17-09 — As an **Owner**, I want to know instantly which packages of a suspect batch have already gone out and to whom, so that a recall takes minutes.

Acceptance: from the batch, a package list shows every unit with its status, and for dispatched units the customer, challan, invoice, vehicle and date; the list exports to PDF and can be sent on WhatsApp to the sales team.

US-M17-10 — As a **Store Keeper**, I want physical stock taking done by scanning, so that the annual count stops taking three days.

Acceptance: a count session records each scanned unit once, then produces a variance report of expected against found — listing units expected in this location but not scanned, units scanned here but expected elsewhere, and unknown codes — which posts as a stock adjustment after approval.

Fields and validation — Packing Unit

Field	Type	Req.	Validation rule
Unit Code	Auto	Yes	10 characters, Crockford Base32, randomly allocated, last character a check digit. Unique across the installation for the life of the database. Never reused, never edited.
Unit Number	Auto	Yes	Human-readable series <code>PU/{FY}/{SEQ:6}</code> . Used in reports and conversation; the QR never carries it.
Packing Level	Select	Yes	0 to 4 per Figure 10.4. Only levels enabled in the industry profile are offered.
Container Type	Select	Yes	Box / Carton / Drum / Bag / Bundle / Roll / Crate / Pallet / Container. The list is per industry profile and per level.
Parent Unit	Select / Scan	No	Must be a higher level than this unit, must not be dispatched, and must not create a loop. Maximum nesting depth 5.
Item	Select	Cond.	Required when the unit holds items directly. Must be active and stockable.
Batch / Serial	Select	Cond.	Required for tracked items. The batch must exist, belong to that item, and have a positive balance at pack time.
Quantity	Decimal	Yes	> 0, maximum 4 decimals. Total packed against a batch may not exceed that batch's produced or received quantity.
Secondary Quantity	Decimal	Cond.	For dual-UOM items (§4.5) — pieces against kilograms. Must fall inside the item's configured conversion tolerance.
Net / Gross / Tare Weight	Decimal	Cond.	≥ 0. Gross must equal net plus tare within a 1% tolerance. Required where the industry profile marks weight as captured per package.
Package n of N	Integer	Yes	Both > 0 and $n \leq N$. Assigned by the system during a bulk pack run.
Source Work Order / GRN	Select	Cond.	One of the two is required. Establishes whether the package came from production or from purchase.

Field	Type	Req.	Validation rule
Warehouse / Location	Select	Yes	Must be inside the user's allowed warehouses. Updated by every transfer scan.
Reserved for Sales Order	Select	No	Order must be Confirmed and must contain this item with a pending quantity. Setting it makes the unit unavailable to other orders.
Status	Select	Yes	Draft / Packed / In Stock / Allocated / Loaded / Dispatched / Delivered / Returned / Opened / Voided. Only the transitions in the status table are permitted.
QC Status	Derived	Yes	Read from the batch's inspection result. Never set directly on the package.
Manufacturing / Expiry Date	Date	Cond.	Inherited from the batch. Expiry must be after manufacturing. Required where the industry profile enforces shelf life.
Label Template	Select	Yes	Must be active and must match the label stock size configured for this container type.
Print Count / Last Printed	Auto	Yes	Incremented by every print. A count above 1 forces the REPRINT mark onto the label.
Voided / Replaced By	Flag / Link	No	Set only by the replace-label action. A voided code can never be scanned into an action again, only viewed.
Remarks	Text	No	Maximum 250 characters. Prints on the label only if the template includes the block.

Status transitions

From	Allowed next	Trigger
Draft	Packed, Voided	Contents confirmed and the label printed.
Packed	In Stock, Opened, Voided	Stock-in scan at the receiving location.
In Stock	Allocated, Loaded, Opened, In Stock (transfer)	Reservation against an order, a transfer scan, or direct loading.
Allocated	Loaded, In Stock (release)	Loading scan, or release when the order is cancelled or amended.
Loaded	Dispatched, In Stock (unload)	Challan confirmation, or reversal before the vehicle leaves.
Dispatched	Delivered, Returned	Proof of delivery upload, or a sales return.
Returned	In Stock, Opened	Only after re-inspection clears it out of quarantine.
Opened / Voided	— terminal —	The code is retired. It remains scannable for enquiry forever.

Business rules

#	Rule
BR-41	A unit code is globally unique and permanent. It is never reused after the package is opened, consumed, voided or delivered.
BR-42	The packing unit never holds an authoritative quantity of its own. Its quantity is a dimension recorded on stock ledger entries; a nightly job reconciles the two and reports any divergence as a defect, not as a stock adjustment.
BR-43	The sum of quantities packed against a batch may never exceed the quantity produced or received for that batch.
BR-44	A package cannot be dispatched while its batch is on QC hold or rejected. This block has no override at any permission level.

#	Rule
BR-45	Scanning is an input method, not an authorisation. Every scanned action passes through exactly the same permission, record-scope and validation checks as the equivalent action typed into a form.
BR-46	Every scan is logged — successes, refusals and duplicates alike — with user, device, timestamp, location where available, and the reason for any refusal.
BR-47	A reprint always reuses the existing code. Generating a new code for an existing package is prohibited; the replace-label action exists for that purpose and voids the old code explicitly.
BR-48	An action on a parent unit applies to every descendant, is previewed before commitment, and is all-or-nothing. A single failing child aborts the whole action and is named.
BR-49	Nesting depth is limited to five levels and a unit can never become its own ancestor. Both are enforced in the database write, not only in the interface.
BR-50	Opening or splitting a unit changes no total stock quantity. Any operation that would change quantity must be posted as a separate, visible adjustment with a reason.
BR-51	A returned package enters a quarantine location and can only reach saleable stock through an inspection that passes.
BR-52	Two devices scanning the same code at the same moment must produce exactly one successful action. The status transition is performed under a row lock, and the loser is told which user and which document took it.
BR-53	Queued offline scans are applied in their original scan order, not their upload order, and are never silently discarded.
BR-54	The public customer scan page exposes only item, batch, dates, standard and the certificate of analysis. It is rate-limited, exposes no list or search endpoint, and can be switched off entirely per installation.
BR-55	Packing units are soft-deleted like every other record. A code that once existed always resolves to something, even if only to "this package was cancelled on this date by this user".
BR-56	The scan resolution endpoint must answer from a single indexed lookup on the code column. It is the most frequently called endpoint in the system and runs on shared hosting.

Database tables introduced

Table	Holds
<code>packing_units</code>	One row per physical package: code (unique index), number, level, container type, parent, warehouse, status, weights, reservation, print count, void chain.
<code>packing_unit_items</code>	Contents of a unit — item, batch or serial, quantity, secondary quantity. A unit may have several lines for a mixed package.
<code>packing_unit_scans</code>	Every scan attempt: unit, user, device, action attempted, result, refusal reason, timestamp, geolocation where permitted, offline flag and original scan time.
<code>packing_unit_events</code>	Status changes, nesting and de-nesting, splits, opens, reprints and label replacements, with before and after values.
<code>label_templates</code>	Template definitions — physical size, blocks and their positions, per container type and per industry profile.
<code>stock_ledger_entries</code>	Existing table from M08, extended with a nullable <code>packing_unit_id</code> column and an index on it.

Interfaces with other modules

Module	Relationship
M03 — Item Master	Supplies default packing configuration per item: standard quantity per box, container type, label template, whether weight is captured per package.

Module	Relationship
M08 — Inventory	Owns the quantity. Packing units add the container dimension to ledger entries and a package-level view of the same stock.
M09 — Production	Packing follows the production entry and is the point at which finished goods become individually identifiable.
M10 — Quality Control	Batch inspection status is read at every dispatch scan. Certificates of analysis are produced per package.
M12 — Dispatch	The packing list is now built from scanned units rather than typed. Package count on the challan is derived, not entered.
M13 — Accounts	Invoice package counts and weights come from the scanned units, which is what the e-way bill and the transporter's receipt must agree with.
§12.3 — Traceability	Traceability moves from batch level to package level: not "which customers got this batch" but "which box went where, and when was it scanned".

Scope discipline. Full GS1 compliance — GTIN, SSCC and GS1 Digital Link — is out of scope for version 1 and should stay out until a customer's own buyer demands it. What version 1 must do is leave room for it: keep a GTIN field on the item master and an SSCC field on the packing unit, both unused. Retrofitting those two columns later is an afternoon; retrofitting a coding scheme that assumed they would never exist is a migration across every label ever printed.

11. Data Model

11.1 Core entity relationships

The diagram shows the central tables and how they connect. Header-and-line pairs (order and order items) are shown as a single box for readability; every transaction table follows this pattern.

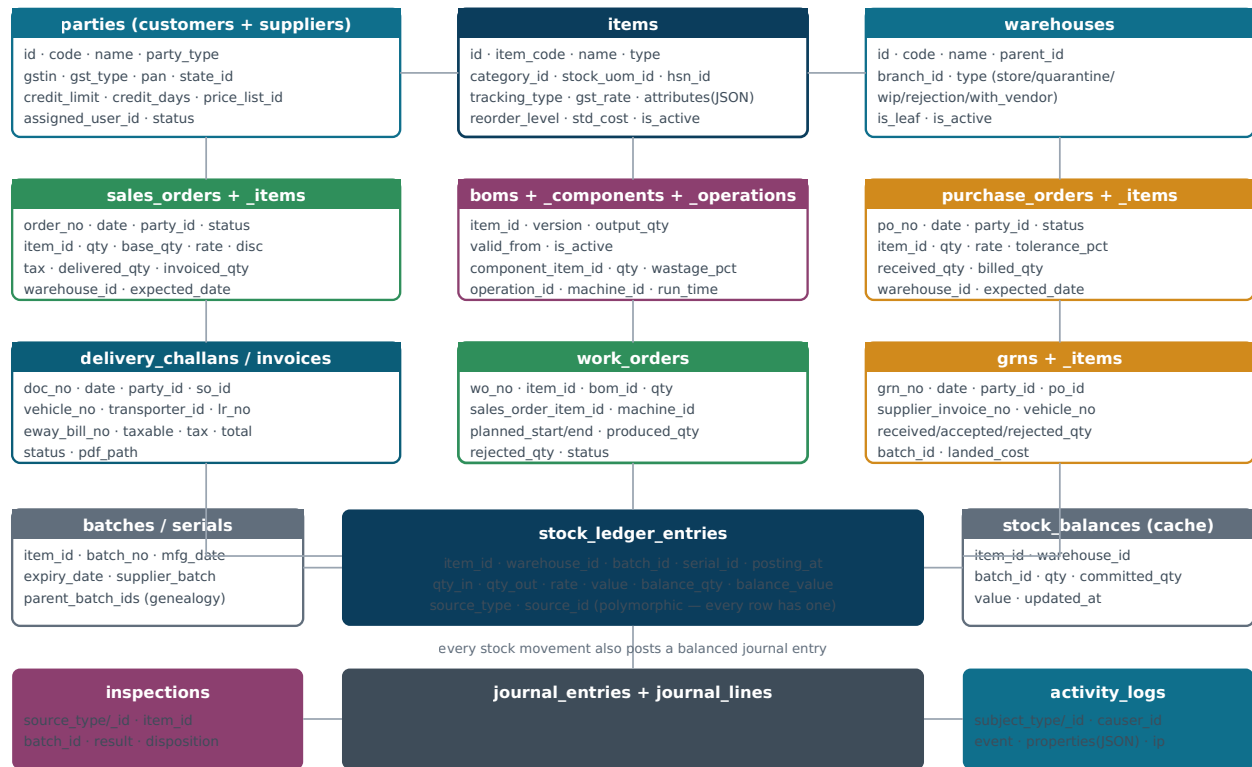


Figure 11.1 — Core data model. The stock ledger sits at the centre; every physical movement passes through it.

11.2 Database conventions

Convention	Rule
Table names	Plural, lower snake case, prefixed by module where helpful: <code>sales_orders</code> , <code>sales_order_items</code> , <code>stock_ledger_entries</code> .
Primary key	<code>id</code> , unsigned big integer, auto-increment.
Foreign keys	<code>{singular_table}_id</code> with a real foreign key constraint, <code>ON DELETE RESTRICT</code> for masters and <code>ON DELETE CASCADE</code> for lines belonging to a header.
Timestamps	<code>created_at</code> , <code>updated_at</code> , <code>deleted_at</code> on every table, plus <code>created_by</code> and <code>updated_by</code> on transaction tables.
Money and quantity	<code>DECIMAL(18,2)</code> for money, <code>DECIMAL(18,4)</code> for quantity and rate, <code>DECIMAL(9,4)</code> for percentages. Never <code>FLOAT</code> or <code>DOUBLE</code> .
Status columns	Short string values backed by a PHP enum, not integers — a database dump should be readable without a lookup table.
JSON columns	Used only for custom field values, industry attributes and log properties. Anything that must be filtered or joined gets a real column.

Convention	Rule
Mandatory indexes	Every foreign key; (<code>item_id</code> , <code>warehouse_id</code> , <code>posting_at</code>) on the stock ledger; (<code>document_no</code>) unique per series; (<code>party_id</code> , <code>date</code>) on all transaction tables; (<code>subject_type</code> , <code>subject_id</code>) on activity logs.
Character set	<code>utf8mb4</code> with <code>utf8mb4_unicode_ci</code> collation, so Gujarati and Hindi text and emoji are stored safely.
Naming of booleans	Prefixed <code>is_</code> or <code>has_</code> : <code>is_active</code> , <code>has_batch</code> .

12. End-to-End Business Flows

12.1 Order to cash

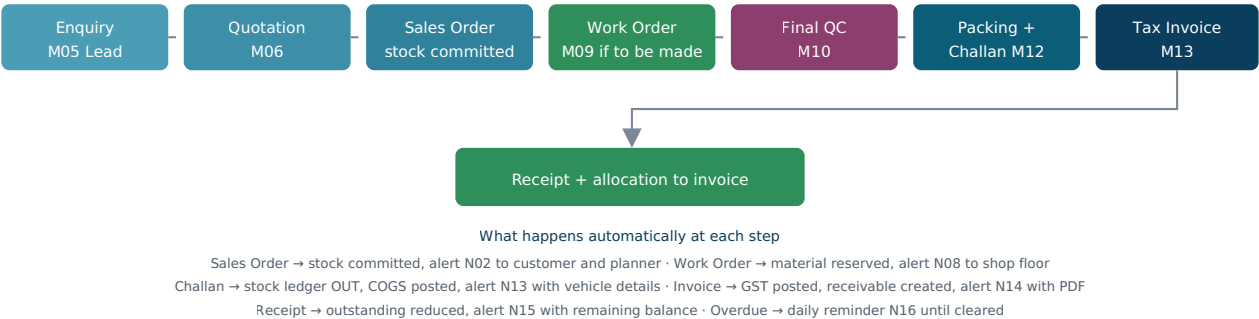


Figure 12.1 — Order to cash, with the automatic postings and alerts at each stage.

12.2 Procure to pay

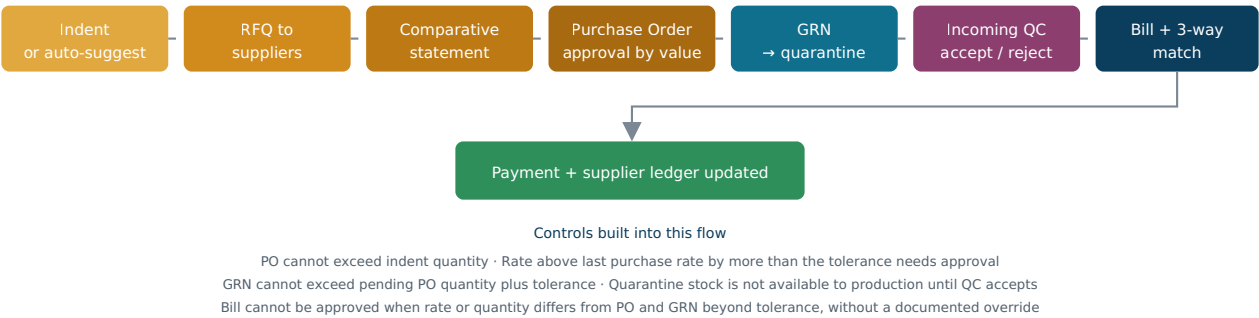


Figure 12.2 — Procure to pay, with the control points that prevent overspending.

12.3 Traceability — the question every customer eventually asks

A customer complains about a batch. Within one screen the ERP must answer: what went into it, who made it, on which machine, who checked it, and where else did that material go? This is achieved by the links already described, walked in both directions.

Direction	Question	Path followed by the system
Backward	"What was this finished batch made from?"	Invoice line → challan line → finished batch → production entry → consumed input batches → GRN → supplier and supplier invoice → incoming inspection result
Forward	"This raw material lot was bad — which customers received it?"	Raw material batch → material issues → work orders → produced batches → challans → invoices → customers, with quantities at every step
Process	"Who and what produced it?"	Produced batch → production entry → machine, mould, operator, shift, date and time, downtime events, in-process inspection results

The traceability report renders this as a tree and exports to PDF, which is often exactly what a customer's quality auditor asks for.

13. Global Validation Standards

These rules apply everywhere and must be implemented once in shared code, not repeated per screen.

13.1 Field-type rules

Data type	Standard rule
Text	Trimmed of leading and trailing spaces; multiple internal spaces collapsed to one; HTML tags stripped unless the field is explicitly rich text; maximum length always enforced at both database and validation level.
Codes	Uppercased automatically; allowed characters restricted; uniqueness checked including soft-deleted rows so a deleted code cannot be silently reused.
Email	RFC-compliant format, lower-cased, maximum 100 characters, with a DNS check available as an optional setting.
Mobile	Non-digits stripped; exactly 10 digits for India with a first digit of 6–9; country code stored separately.
Quantity	Greater than zero on transaction lines; maximum 4 decimals; upper bound of 9,999,999.9999 to catch typing errors such as an extra zero.
Rate and amount	Zero or greater; maximum 4 decimals for rate and 2 for amount; an amount above ₹10 crore on a single line prompts confirmation.
Percentage	Between 0 and 100 unless the field explicitly allows more; 2 decimals.
Date	Valid calendar date; never in a closed financial period; future dates blocked unless the field explicitly allows them; date ranges always validated so the end is not before the start.
File upload	Extension whitelist plus MIME type read from the file's own content; size limit per field; filename sanitised; files stored outside the public web root and served through an authenticated controller.
Dropdown / foreign key	The submitted value must exist, be active, and be inside the user's data scope. Never trust an ID posted from the browser.

13.2 Cross-cutting rules

#	Rule	Detail
V1	Server-side is the truth	Every rule in this document is implemented in a Laravel FormRequest. Browser validation is a convenience only.
V2	One error, one field	Errors are returned per field and displayed beside the field, not as a single combined message at the top.
V3	Messages are in plain language	"Delivery date cannot be before the order date" — not "validation.before".
V4	Documents need at least one line	Enforced on every header-and-line document.
V5	Totals are computed, never trusted	The server recalculates every total from the lines. A tampered total posted from the browser is ignored, not saved.
V6	Concurrency is handled	Every editable record carries a version or updated-at token. Saving a record that changed since it was loaded returns "This record was changed by another user" rather than silently overwriting.
V7	Idempotent submission	Every create form carries a one-time token so a double-click or a browser back-and-resubmit cannot create two documents.
V8	Status governs editability	Each document type declares which statuses are editable. The check lives in one place and is applied by both the UI and the API.

#	Rule	Detail
V9	Deletion is checked for references	Before any soft delete the system counts dependent records and refuses with a message naming what is blocking.
V10	Imports use the same rules	Bulk import passes every row through the same FormRequest as the screen, so imported data can never be worse than typed data.

14. Non-Functional Requirements

14.1 Performance

#	Requirement	Target
P1	Page load — list screens	Under 2 seconds with 100,000 records in the table, using server-side pagination and proper indexes
P2	Page load — document entry screen	Under 1.5 seconds including master dropdowns, which are cached
P3	Save a document with 50 lines	Under 3 seconds including stock ledger and accounting postings
P4	PDF generation	Under 3 seconds for a 100-line invoice
P5	Report rendering	Under 5 seconds for one year of data; longer reports run as queued jobs
P6	Notification dispatch	Queued within 200 ms of the triggering event; delivered within 60 seconds under normal provider response
P7	Concurrent users	25 concurrent active users on entry-level shared hosting; 100 on a 4 vCPU / 8 GB VPS
P8	Database growth	The design must remain performant to 5 million stock ledger rows without partitioning

14.2 Reliability and availability

#	Requirement	Detail
R1	Data integrity	All multi-table writes inside transactions with row-level locking on stock balances and numbering series.
R2	Backup	Automatic daily database and file backup, retained for a configurable number of days, with an off-site copy option to an FTP target or an S3-compatible bucket.
R3	Restore	Documented and tested restore procedure with a target recovery time of under 2 hours and a recovery point of under 24 hours.
R4	Queue resilience	Failed jobs are retried with backoff and land in a failed-jobs table with a one-click retry from the admin screen.
R5	Graceful degradation	If WhatsApp or Firebase is unreachable, the ERP continues to work fully — only the outbound alert is queued and retried.
R6	Maintenance mode	Updates put the application into maintenance mode with a friendly page and an IP allow-list for the implementer.

14.3 Security

#	Requirement	Detail
S1	Transport security	HTTPS enforced; HTTP redirected; HSTS header enabled once the certificate is stable.
S2	Injection protection	Parameter binding everywhere; no string-concatenated SQL; output escaped by default in Blade.
S3	CSRF and clickjacking	CSRF tokens on every state-changing form; <code>X-Frame-Options: SAMEORIGIN</code> ; a strict content security policy.

#	Requirement	Detail
S4	File upload safety	Uploads stored outside the web root, never executed, served through an authenticated controller with the correct content type and a download disposition.
S5	Secrets at rest	Integration tokens and the Firebase service account encrypted using the application key. The <code>.env</code> file is outside the web root and unreadable by other accounts.
S6	Rate limiting	Login, password reset, API endpoints and export routes are throttled per user and per IP.
S7	Dependency hygiene	Composer and npm audits run before each release; no known high-severity vulnerability ships.
S8	Error disclosure	Debug mode is off in production. Users see a reference code; the detail goes to the log.
S9	Personal data	Employee and customer personal data is access-controlled and every view of salary data is logged.

14.4 Usability

#	Requirement	Detail
U1	Keyboard-first data entry	Tab order follows the visual order; a line can be added with a keyboard shortcut; the save shortcut works on every form. Data entry staff should rarely need the mouse.
U2	Search everywhere	A global search finds documents by number, party by name or GSTIN, and items by code, name or barcode.
U3	Consistent screens	Every list screen has the same filter, column-chooser, export and pagination behaviour; every document screen has the same tab layout of Details, Attachments and History.
U4	Responsive	Fully usable on a 10-inch tablet for shop floor and stores; the phone layout covers approvals, dashboards and lookups.
U5	Language	English, Hindi and Gujarati interface with the ability to add more through translation files.
U6	Helpful errors	Every error tells the user what to do next, not only what went wrong.
U7	Number formatting	Indian grouping (1,24,560.00) as the default, switchable to international grouping.

14.5 Maintainability

#	Requirement	Detail
M1	Coding standard	PSR-12 enforced by Laravel Pint in continuous integration.
M2	Static analysis	PHPStan at level 6 or above with no new errors permitted.
M3	Automated tests	Feature tests for every user story that changes data; unit tests for the stock ledger, costing, tax and numbering engines. Target 70% coverage overall and 95% on those four engines.
M4	Migrations	Always reversible, never destructive without an explicit data-migration step. Never edit a migration that has shipped.
M5	Documentation	Every module ships with an admin guide page and a short video; the API is documented with request and response examples.

15. cPanel Deployment Specification

15.1 The challenge in one paragraph

Laravel expects a document root pointing at a `public` folder, a long-running queue worker, a per-minute scheduler, and shell access. Shared cPanel hosting gives none of these by default. This section states exactly how the application must be built and installed so that it works on ordinary cPanel hosting without compromise, and takes advantage of a VPS when one is available.

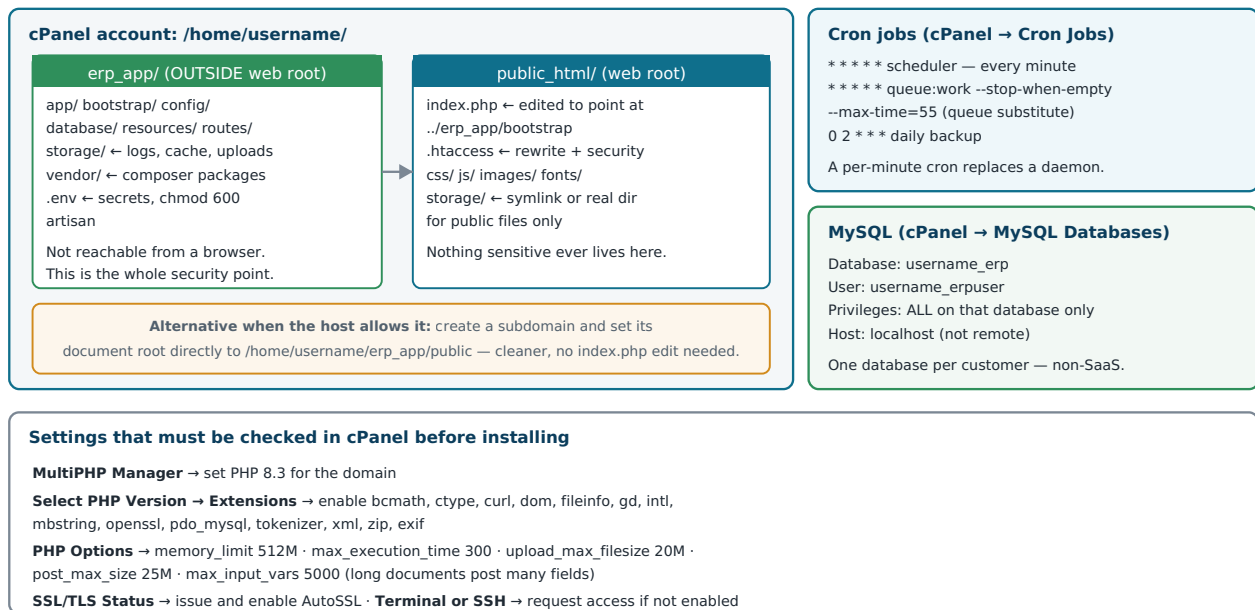


Figure 15.1 — cPanel deployment layout. Application code lives outside the web root; only the public folder is exposed.

15.2 Directory layout and the index.php shim

When the host will not let you re-point a document root, the standard pattern is to keep the framework outside `public_html` and edit the copied `index.php` to look one level up. Two lines change:

```
// public_html/index.php
require __DIR__.'../erp_app/vendor/autoload.php';
$app = require_once __DIR__.'../erp_app/bootstrap/app.php';
```

Everything else in the file stays as Laravel shipped it. The `public` folder's assets are copied or symlinked into `public_html`.

15.3 Required .htaccess

```
# public_html/.htaccess
<IfModule mod_rewrite.c>
  RewriteEngine On
  RewriteRule ^(.*)/$ /$1 [L,R=301]          # trailing slash
  RewriteCond %{REQUEST_FILENAME} !-d
  RewriteCond %{REQUEST_FILENAME} !-f
  RewriteRule ^ index.php [L]
</IfModule>

# Force HTTPS
RewriteCond %{HTTPS} off
RewriteRule ^(.*)$ https://%{HTTP_HOST}%{REQUEST_URI} [L,R=301]

# Block access to sensitive files even if one is copied here by mistake
<FilesMatch "^(.env|git|htaccess)|composer\.(json|lock)|package\.json$">
  Require all denied
</FilesMatch>

# Security headers
<IfModule mod_headers.c>
  Header always set X-Content-Type-Options "nosniff"
  Header always set X-Frame-Options "SAMEORIGIN"
  Header always set Referrer-Policy "strict-origin-when-cross-origin"
</IfModule>

# Compression and caching for assets
<IfModule mod_deflate.c>
  AddOutputFilterByType DEFLATE text/html text/css application/javascript application/json
</IfModule>
<IfModule mod_expires.c>
  ExpiresActive On
  ExpiresByType text/css "access plus 1 year"
  ExpiresByType application/javascript "access plus 1 year"
  ExpiresByType image/png "access plus 1 year"
</IfModule>
```

15.4 Cron jobs — replacing the daemon

Shared hosting will not run a permanent queue worker, and killing long processes is common. The pattern below works reliably: a per-minute cron starts a worker that exits by itself before the next minute begins, so at most one worker runs at a time and nothing is left hanging.

Schedule	Command	Purpose
* * * * *	/usr/local/bin/php /home/user/erp_app/ artisan schedule:run >> /dev/null 2>&1	Drives all scheduled tasks: overdue reminders, daily summary, backups, maintenance-due checks, token cleanup
* * * * *	/usr/local/bin/php /home/user/erp_app/ artisan queue:work -- queue=notifications,default --stop-when- empty --max-time=55 --tries=3 --sleep=3	Processes queued jobs — notifications, PDFs, imports, report generation — then exits cleanly
0 2 * * *	/usr/local/bin/php /home/user/erp_app/ artisan erp:backup --database --files	Nightly backup with retention pruning
*/10 * * * *	/usr/local/bin/php /home/user/erp_app/ artisan erp:health-check	Writes a heartbeat that the admin health panel reads; alerts if the scheduler itself has stopped

Two things that catch people out. First, use the absolute path to the PHP CLI binary — cPanel's default `php` in cron is often an older version than the one selected for the website; check with `which php83` or use the EasyApache path such as `/opt/cpanel/ea-php83/root/usr/bin/php`. Second, add `--stop-when-empty` and `--max-time`, otherwise overlapping workers accumulate and the host suspends the account for excessive processes.

15.5 Configuration for shared hosting

Setting	Shared cPanel	VPS or dedicated
<code>QUEUE_CONNECTION</code>	<code>database</code>	<code>redis</code>
<code>CACHE_STORE</code>	<code>file</code> or <code>database</code>	<code>redis</code>
<code>SESSION_DRIVER</code>	<code>file</code>	<code>redis</code>
<code>PDF_ENGINE</code>	<code>dompdf</code> (pure PHP, no binary needed)	<code>chromium</code> for better CSS support
<code>APP_DEBUG</code>	<code>false</code> — always	<code>false</code>
<code>LOG_CHANNEL</code>	<code>daily</code> with 14-day retention, so logs never fill the disk quota	<code>daily</code> or a log service
OPcache	Enable through Select PHP Version if available	Enable with a 256 MB buffer
Asset build	Build locally and ship compiled assets — Node is usually unavailable on shared hosting	Build on the server or in CI

15.6 Installation checklist

Step	Action	Verification
1	Confirm PHP 8.3 and all extensions in MultiPHP Manager	The installer's requirement screen shows all green
2	Create the database and a user, grant all privileges on that database	Test connection from the installer
3	Upload the release ZIP and extract to <code>/home/user/erp_app</code>	Folder structure matches Figure 15.1
4	Copy the contents of <code>erp_app/public</code> into <code>public_html</code> and edit <code>index.php</code>	The domain loads the installer page
5	Set permissions: folders 755, files 644, <code>storage</code> and <code>bootstrap/cache</code> writable (775), <code>.env</code> 600	The installer's permission check passes
6	Run the web installer: database credentials, company details, industry profile, admin user	Migrations and seeders complete; login screen appears
7	Add the four cron jobs	The health panel shows the scheduler heartbeat within two minutes
8	Enable AutoSSL and confirm HTTPS redirect	The padlock appears and HTTP redirects to HTTPS

Step	Action	Verification
9	Configure WhatsApp and Firebase credentials, then use the test buttons	Test message and test push both arrive
10	Run <code>config:cache</code> , <code>route:cache</code> and <code>view:cache</code>	Page load times drop noticeably
11	Import opening masters and opening stock, then lock the opening date	Stock valuation matches the customer's opening figures
12	Trigger a manual backup and download it	The archive restores successfully in a test environment

15.7 Known cPanel constraints and how the product handles them

Constraint	Product response
No SSH on some plans	A web-based installer and updater performs everything artisan would do — migrations, cache clearing, symlink creation — through the browser, protected by a one-time installation key.
<code>symlink()</code> disabled	The installer detects this and falls back to a real directory plus a controller route that serves public files, so uploads still work.
Process and CPU limits	Heavy work — imports, bulk PDFs, large reports, ledger rebuilds — always runs in the queue in small chunks, never in a web request.
Inode limits	Log rotation capped at 14 days; cache files pruned nightly; generated PDFs older than a configurable period are archived into monthly ZIP files.
No Redis	Database and file drivers are the default and are tested as first-class options, not as a degraded fallback.
Shared MySQL performance	Query design assumes no query cache; heavy reports use summary tables refreshed by the scheduler rather than aggregating raw rows on every view.
Email deliverability	The default is SMTP through an authenticated provider rather than the server's <code>mail()</code> function, because shared-host IP addresses are frequently blacklisted.

16. Implementation Roadmap

Phase	Duration	Delivers	Milestone / exit criteria
0	2 weeks	Project setup: repository, coding standards, continuous integration, base Laravel skeleton, layered structure, base model with logging, numbering engine, permission scaffolding	A "hello module" can be created end to end with logging, permissions and PDF working
1	6 weeks	M01 Company and masters · M02 Users and RBAC · M03 Item master · M08 Inventory core with stock ledger · M16 Settings foundation · installer	Opening stock can be loaded and a stock report tallies. This is the foundation everything else stands on.
2	8 weeks	M06 Sales · M07 Purchase · M12 Dispatch · M04 BOM · M09 Production · PDF engine · core reports	A full order-to-cash and procure-to-pay cycle runs, with stock moving correctly at every step
3	6 weeks	M13 Accounts and GST · M10 Quality · M11 Maintenance · M05 CRM · notification engine with WhatsApp and Firebase	Books tally with stock; GSTR-1 working sheet matches the sales register; alerts are delivered reliably
4	5 weeks	M14 HR and payroll · M15 full report and dashboard set · customization framework completed · customer portal	All 16 modules feature-complete against this document
5	4 weeks	Industry profile packs for all eleven industries · performance tuning · security review · documentation and training videos · pilot installation	A pilot customer runs a full month on the system with no parallel manual records

Durations assume a team of three developers, one QA engineer and one implementation consultant. Phases 2 and 3 overlap partially: accounts posting rules are written alongside the sales and purchase modules rather than after them, because retrofitting accounting into finished documents is far more expensive than building it in.

16.1 Acceptance approach

Stage	What happens
Story acceptance	Each user story in this document is a test case. QA verifies the acceptance criteria exactly as written.
Module acceptance	The business owner runs their own real transactions through the module and signs off against the field and validation tables.
Integration acceptance	End-to-end scenarios from Section 12 are run with real data, checking that stock, accounting and alerts all agree.
Parallel run	One month of running the ERP alongside the existing method, reconciling stock and sales figures at the end of the month.
Go-live	Opening balances locked, manual records stopped, support arrangement in place for the first quarter.

17. Appendices

17.1 Appendix A — Document status codes

Document	Statuses in order
Lead	New → Contacted → Qualified → Quoted → Negotiation → Won / Lost
Quotation	Draft → Sent → Accepted / Rejected / Expired → Converted
Sales Order	Draft → Pending Approval → Confirmed → In Production → Ready → Partially Delivered → Delivered → Invoiced → Closed; plus Cancelled and On Hold
Delivery Challan	Draft → Dispatched → Delivered → Invoiced; plus Cancelled
Invoice	Draft → Confirmed → Partially Paid → Paid; plus Cancelled
Purchase Indent	Draft → Pending Approval → Approved → Partially Ordered → Ordered → Closed; plus Rejected and Cancelled
Purchase Order	Draft → Pending Approval → Approved → Sent → Partially Received → Received → Closed; plus Cancelled
GRN	Draft → Posted → Under Inspection → Accepted / Partially Accepted / Rejected
Work Order	Draft → Released → In Progress → Completed → Closed; plus Cancelled and On Hold
Inspection	Pending → In Progress → Passed / Failed / Accepted with Deviation
Voucher	Draft → Confirmed → Reconciled; plus Cancelled (by reversal only)

17.2 Appendix B — Sample permission matrix

Permission group	Store Keeper	Purchase Officer	Sales Exec	Production Planner	Accountant	Management
Item master	view	view, create	view	view	view	view
Party master	view	full (supplier)	full (customer, own)	view	view, update	view
Sales order	view	—	create, update, view	view (no rates)	view	view, approve
Purchase order	view	create, update, print	—	view	view	view, approve
GRN	create, update, print	view	—	view	view	view
Material issue	create, update	—	—	view	—	view
Work order	view	—	view	full	—	view
Production entry	—	—	—	view, update	—	view
Invoice	—	—	view	—	full	view
Payments / receipts	—	—	—	—	full	view, approve
Reports	stock only	purchase, stock	sales (own)	production, stock	finance, sales, purchase	all

Permission group	Store Keeper	Purchase Officer	Sales Exec	Production Planner	Accountant	Management
Settings	—	—	—	—	—	view

This is a starting point supplied as a seeder. Every customer adjusts it, and every adjustment is logged.

17.3 Appendix C — Numbering series defaults

Document	Pattern	Example
Quotation	QT/{FY}/{SEQ:5}	QT/26-27/00013
Sales Order	SO/{FY}/{SEQ:5}	SO/26-27/00042
Delivery Challan	DC/{FY}/{SEQ:5}	DC/26-27/00097
Tax Invoice	INV/{FY}/{SEQ:5}	INV/26-27/00187
Purchase Order	PO/{FY}/{SEQ:5}	PO/26-27/00214
GRN	GRN/{FY}/{SEQ:5}	GRN/26-27/00331
Work Order	WO/{FY}/{SEQ:5}	WO/26-27/00508
Material Issue	MI/{FY}/{SEQ:5}	MI/26-27/01120
Inspection	QC/{FY}/{SEQ:5}	QC/26-27/00276
Payment Voucher	PV/{FY}/{SEQ:5}	PV/26-27/00088
Packing Unit	PU/{FY}/{SEQ:6}	PU/26-27/004128
Pallet / handling unit	PL/{FY}/{SEQ:5}	PL/26-27/00219
QR scan code	Random Base32, 10 chars	7K4M9XQ2PB (not a series — never sequential)
Batch (auto)	{ITEM}/{YY}{MM}/{SEQ:3}	PVC110/2607/014

17.4 Appendix D — Open questions to settle before build starts

#	Question	Why it matters
Q1	Which industry is the pilot customer?	The first industry profile gets built and validated first; the rest follow the same pattern.
Q2	Is multi-branch needed at version 1, or one plant only?	Branch scoping touches every query. Adding it later is expensive; leaving it in when unused costs little.
Q3	Is e-invoicing (IRN) required, and through which GSP?	The GSP contract has a lead time and defines the payload format.
Q4	Which biometric attendance device is in use?	Determines the punch file format for the importer.
Q5	Will accounting stay in Tally, or move fully into the ERP?	If Tally stays, an export bridge is needed instead of the full finance module — a significant scope difference.
Q6	Is a native mobile app required, or is a responsive web app sufficient?	A Flutter client is a separate project with its own API surface and release cycle.
Q7	What is the expected number of stock transactions per day?	Determines whether shared hosting is adequate or a VPS is required from day one.

End of specification. Any change to the requirements in this document after baseline approval should be raised as a written change request, assessed for effort and impact on the roadmap, and approved by the business owner and the project manager before development begins.